



IMT Atlantique

Bretagne-Pays de la Loire
École Mines-Télécom



Hybridizing Metaheuristics with Machine Learning for Combinatorial Optimization: A taxonomy and Learning to Select Operators

Maryam KARIMI MAMAGHAN

To obtain the PhD degree

Reviewers:

**Jin-Kao HAO
Kenneth SORRENSEN**

Jury members: Olga BATAIA

**Axel PARMENTIER
Sébastien VEREL**

Director:

Patrick MEYER

Supervisors:

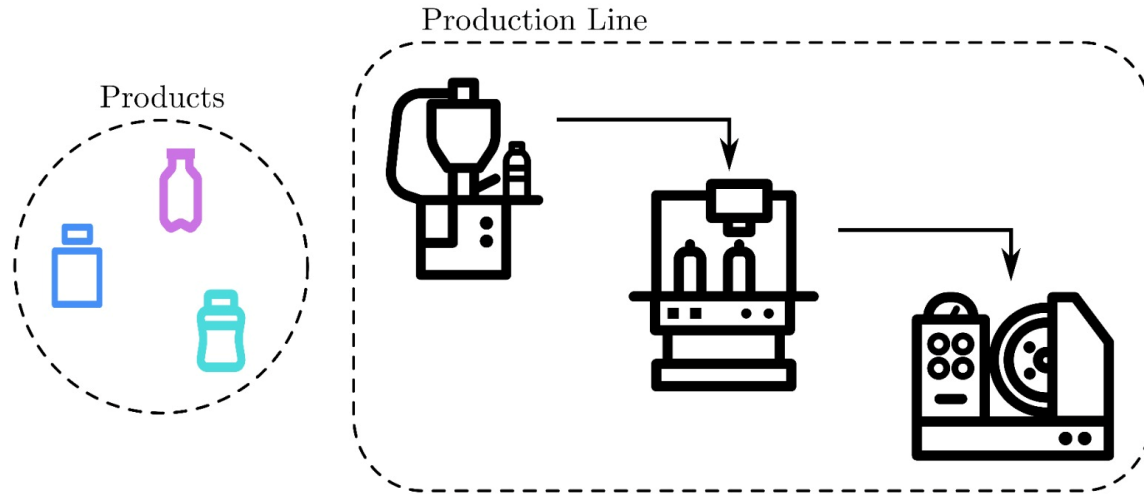
**Mehrdad MOHAMMADI
Bastien PASDELOUP**

July 20, Brest, France

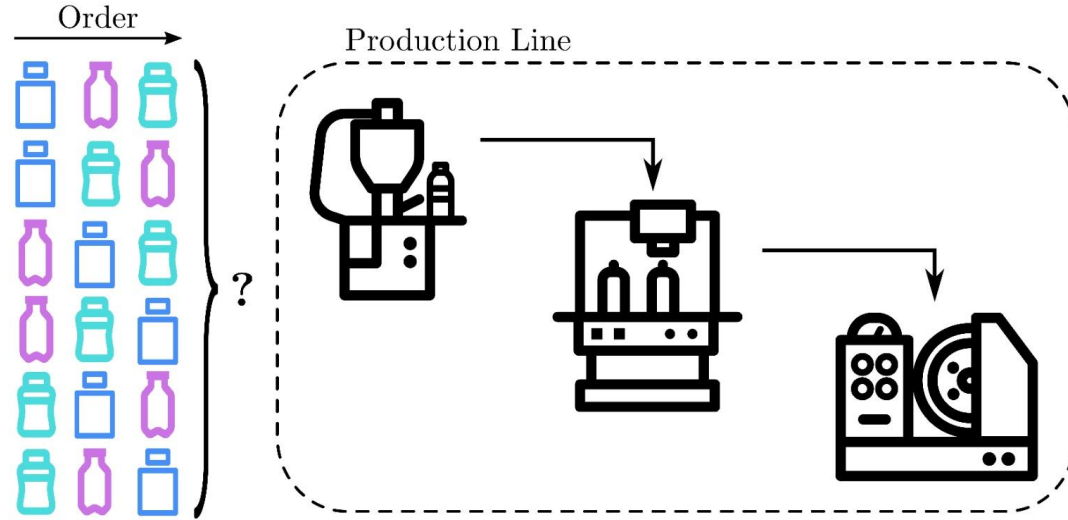
- Introduction
- Research questions and contributions
- The proposed framework
- Experimental design
- Numerical results
- Discussion and conclusion
- Future research

Introduction

A set of products, a set of machines, the processing time of products on machines

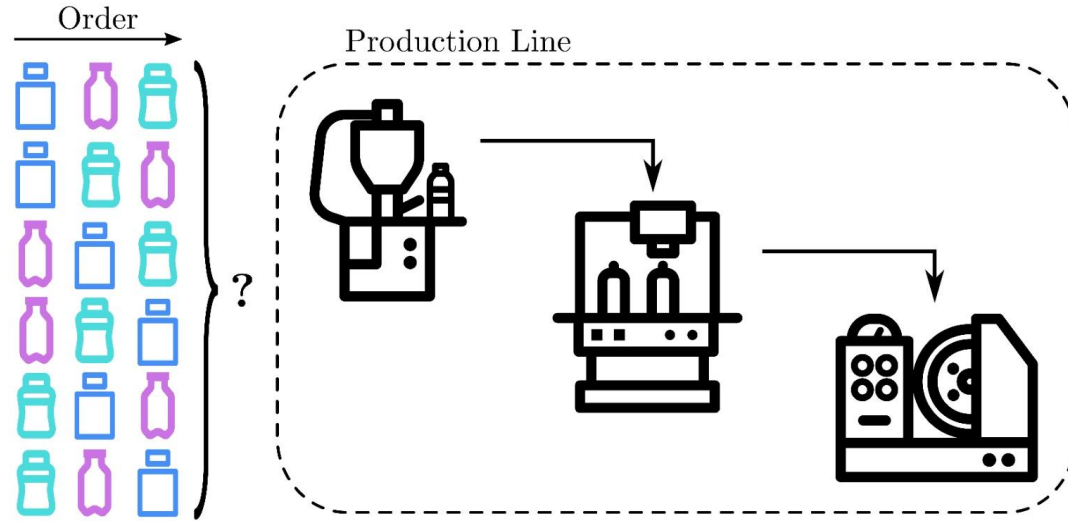


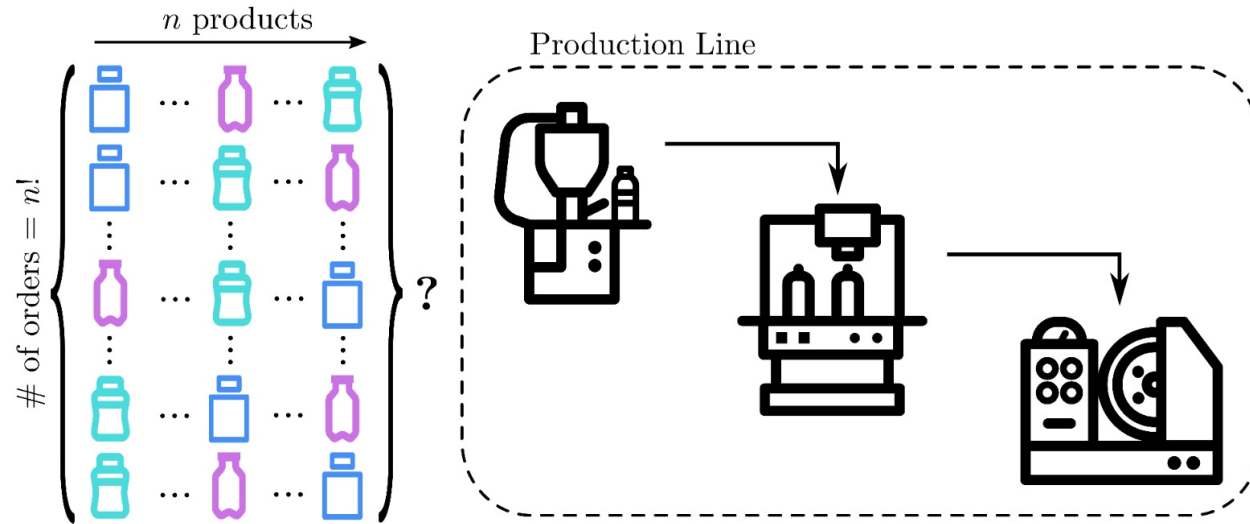
Finding the order of products/jobs to be processed on machines to achieve a specific objective



Finding the order of products/jobs to be processed on machines to achieve a specific objective

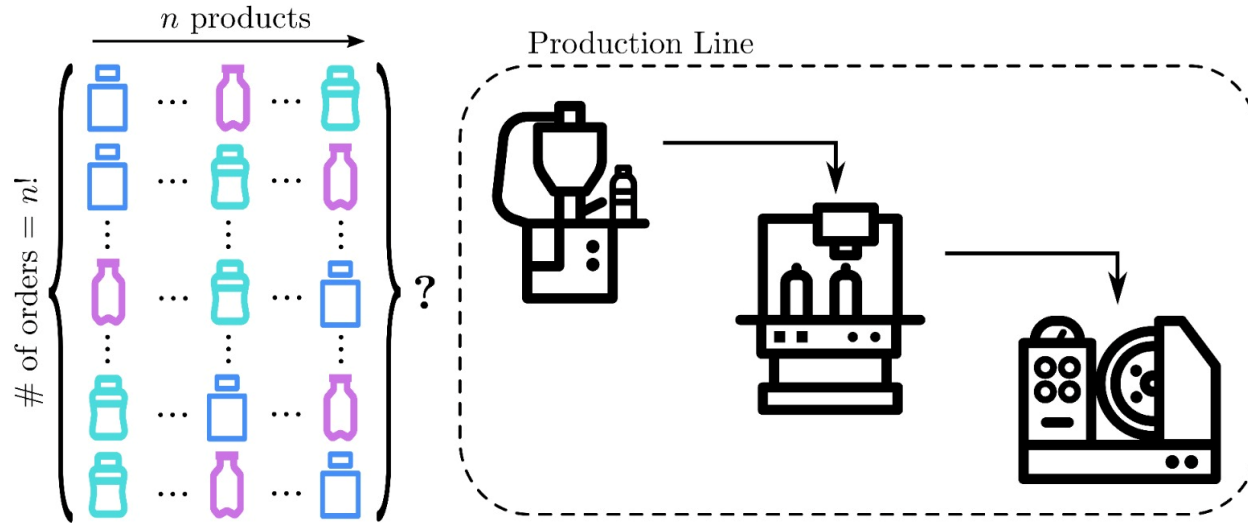
Production scheduling problem (permutation flowshop scheduling problem-PFSP)





With n number of products

$n=3 \rightarrow 6$ solutions

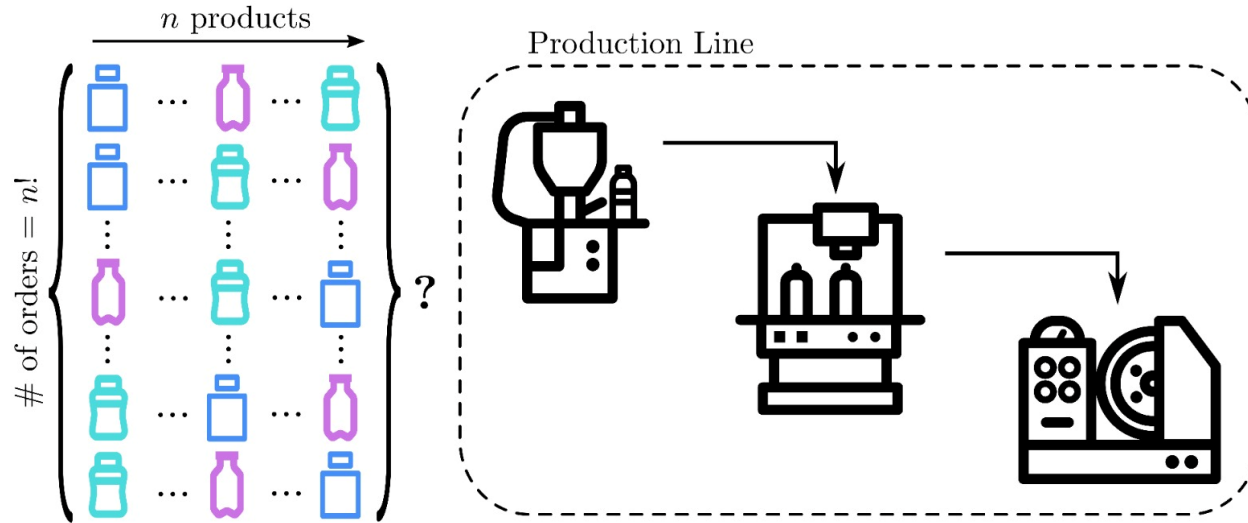


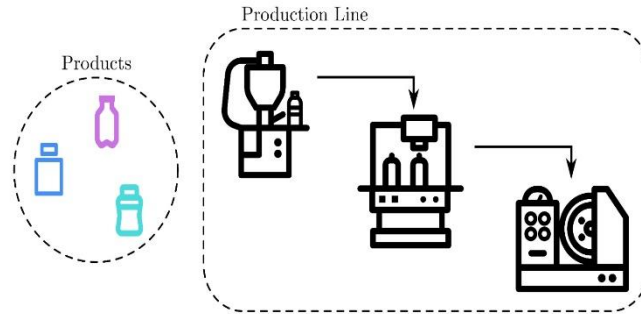
With n number of products

$n=3$ → 6 solutions

$n=10$ → $3.6E+6$ solutions

$n=15$ → $1.3E+12$ solutions

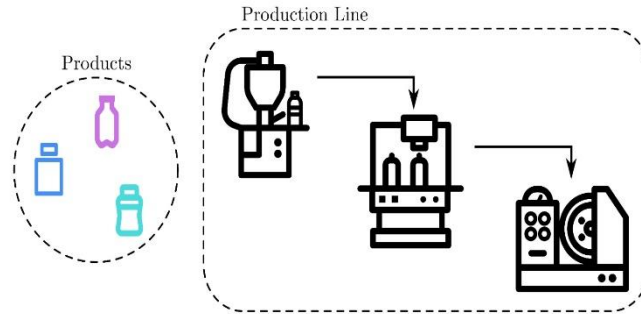




Permutation flowshop Scheduling
Problem



Vehicle Routing Problem



Permutation flowshop Scheduling Problem

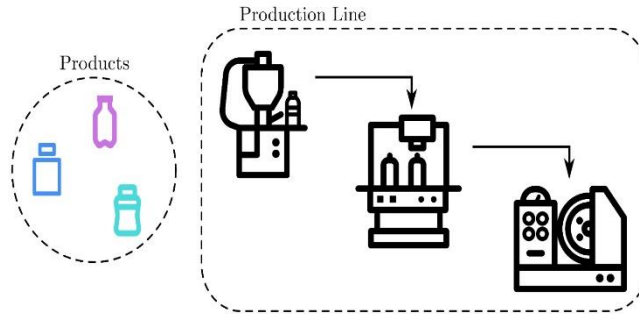


Traveling Salesman Problem

We develop an algorithm to find the solutions to these problems efficiently



Vehicle Routing Problem



Permutation flowshop Scheduling Problem



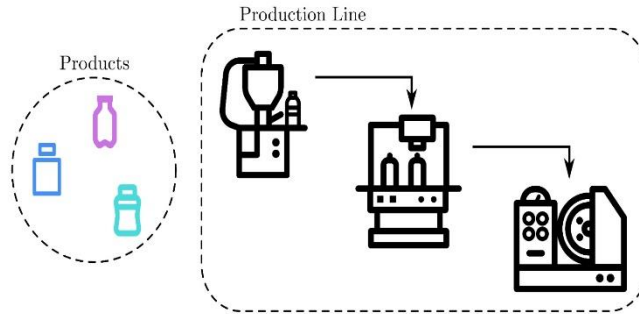
Traveling Salesman Problem

We develop an algorithm to find the solutions to these problems efficiently

Combinatorial optimization problems



Vehicle Routing Problem



Permutation flowshop Scheduling Problem



Traveling Salesman Problem

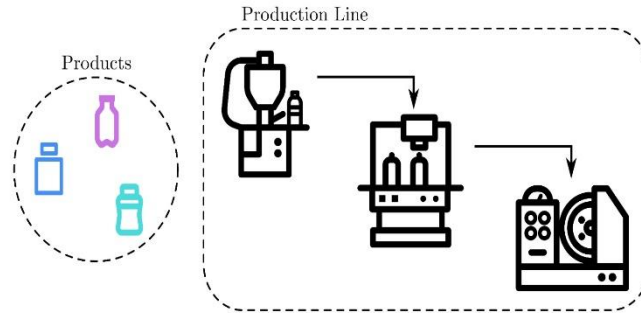
Combinatorial optimization problems

8

- ▶ Discrete decision variables
- ▶ Finite (usually huge) solution space
- ▶ Mostly NP-hard



Vehicle Routing Problem



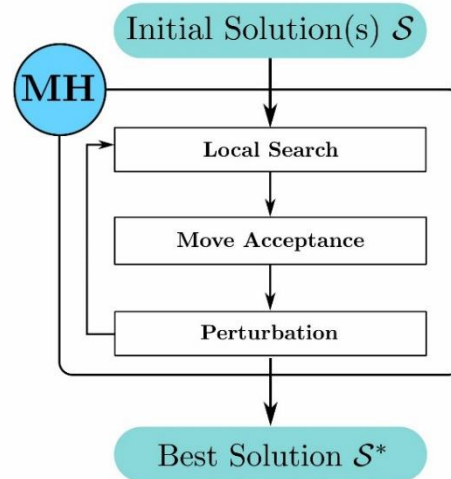
Permutation flowshop Scheduling
Problem



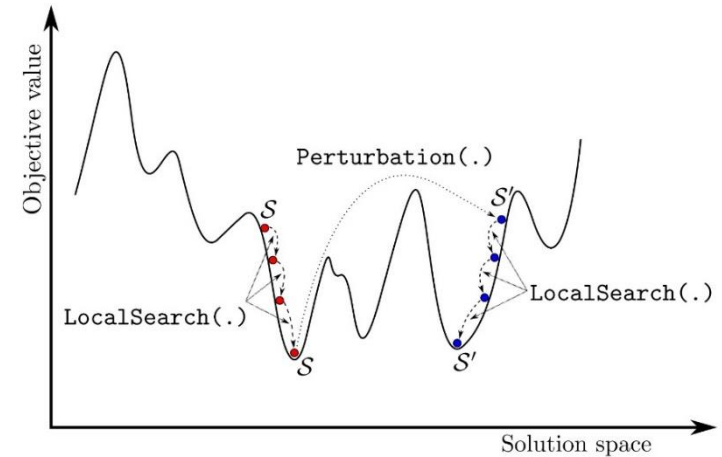
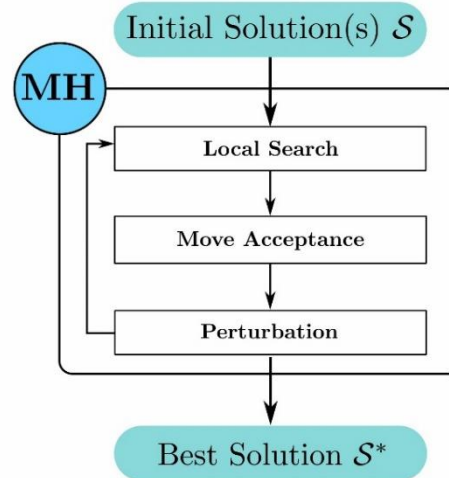
Traveling Salesman Problem

- ▶ Approximation algorithms
- ▶ Finding good solutions in reasonable time
- ▶ No guarantee for optimality

- ▶ Approximation algorithms
- ▶ Finding good solutions in reasonable time
- ▶ No guarantee for optimality



- ▶ Approximation algorithms
- ▶ Finding good solutions in reasonable time
- ▶ No guarantee for optimality



- ▶ Subfield of artificial intelligence
- ▶ ML algorithms learn how to perform a task or make predictions using data
- ▶ Applications:
 - Automatic object detection
 - Speech recognition
 - Predictive analytics for forecasting purposes
 - Optimization

Supervised learning

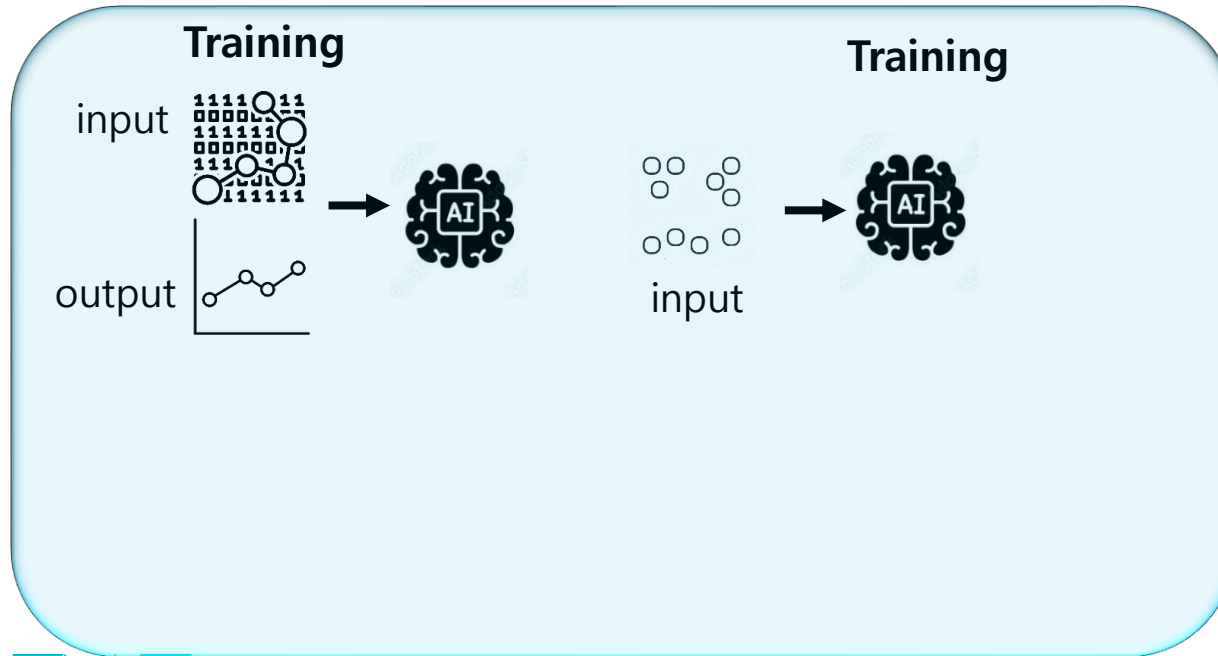
Unsupervised learning

Reinforcement learning

Supervised learning

Unsupervised learning

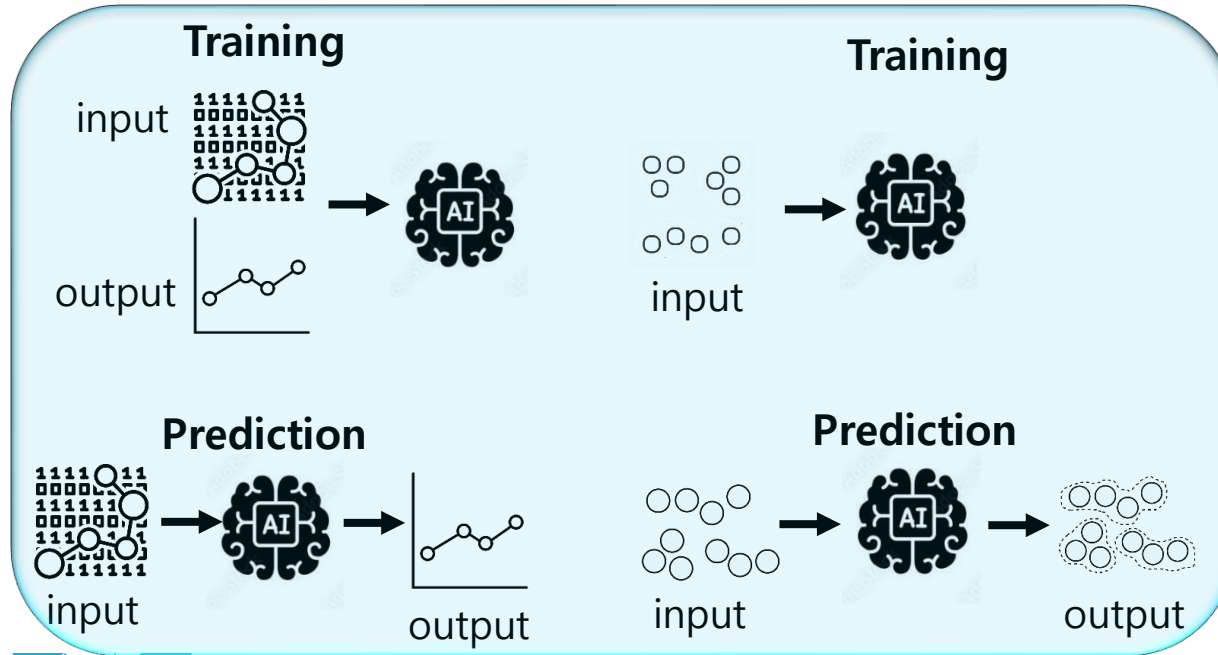
Reinforcement learning



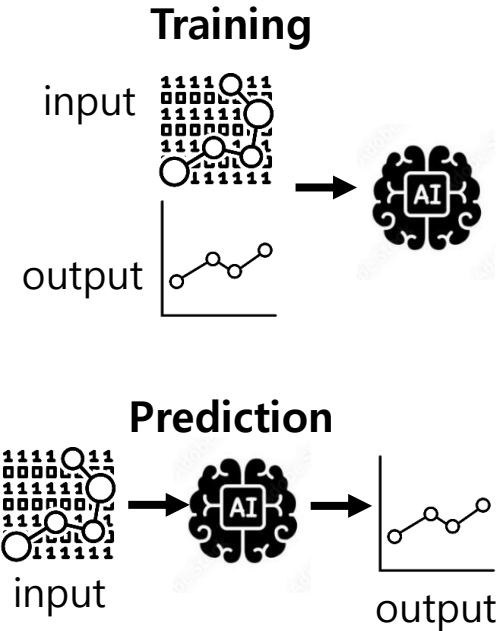
Supervised learning

Unsupervised learning

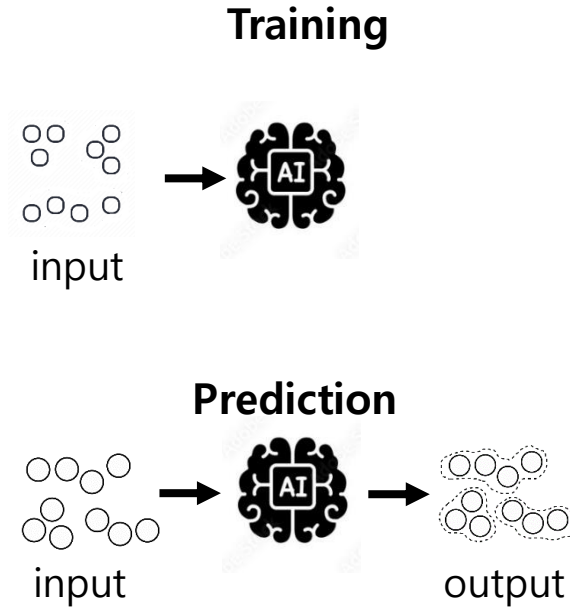
Reinforcement learning



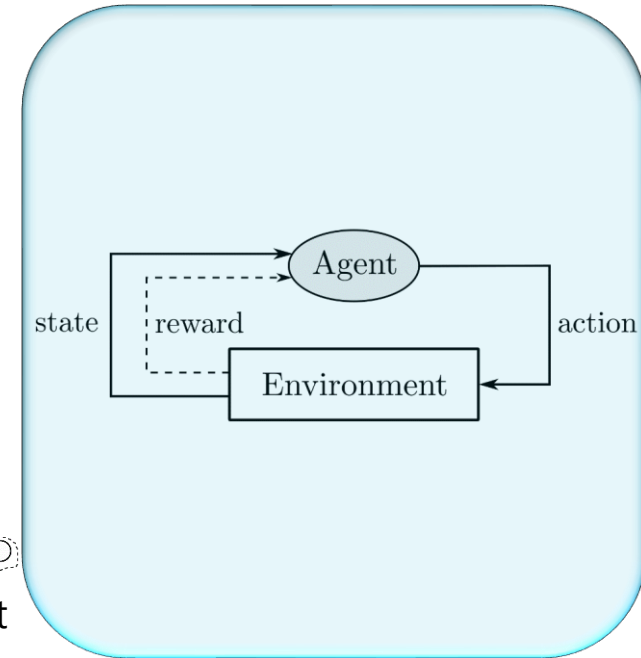
Supervised learning



Unsupervised learning



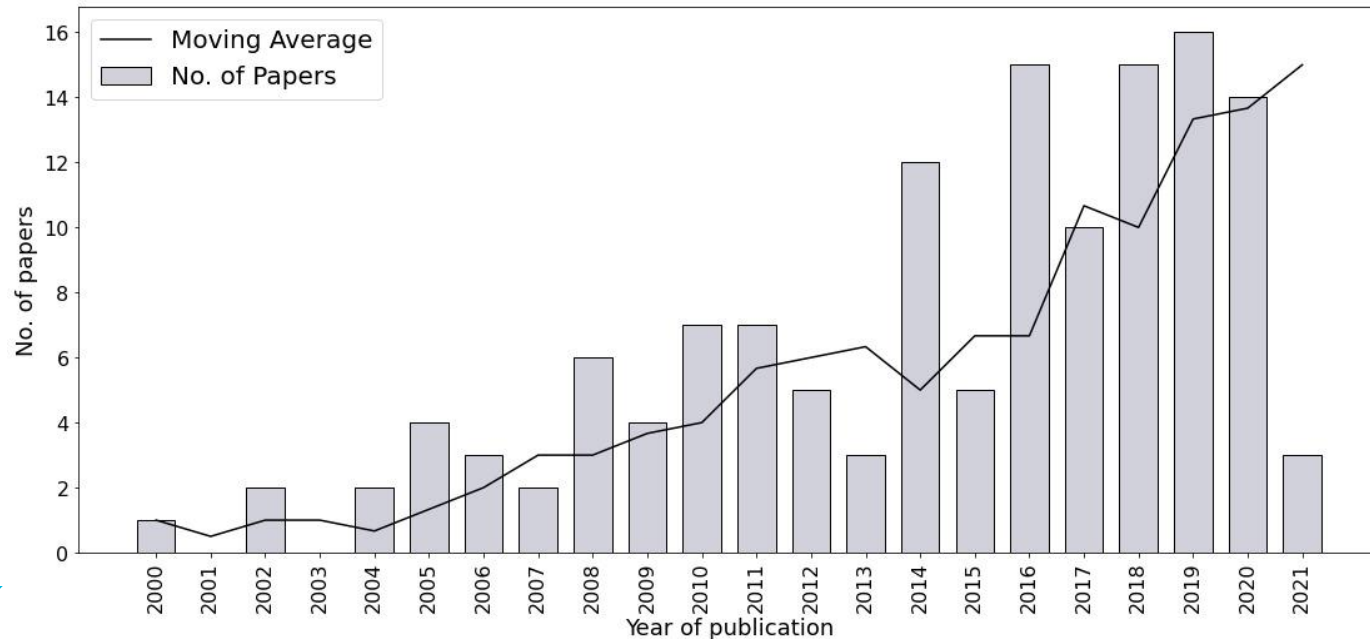
Reinforcement learning



Research Questions and Contributions

Q1. How and for which purposes machine learning can help meta-heuristics?

Q1. How and for which purposes machine learning can help meta-heuristics?



Q1. How and for which purposes machine learning can help meta-heuristics?

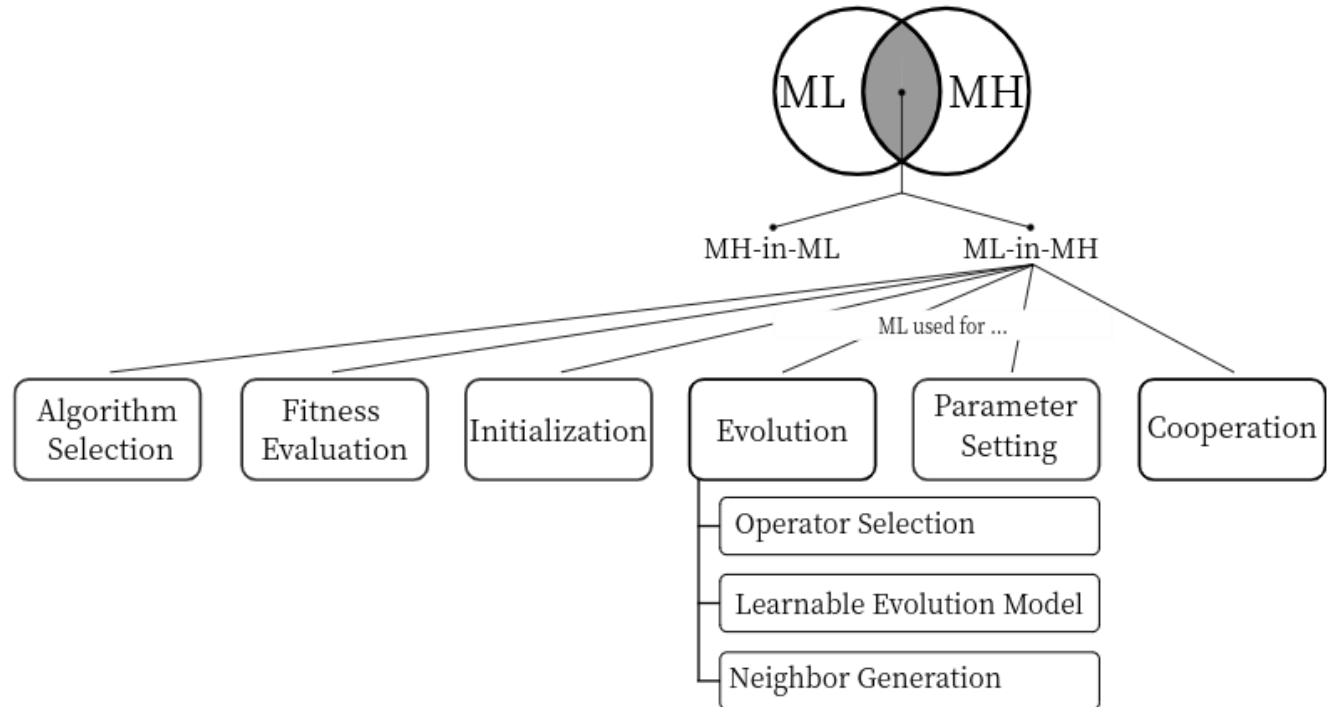
Existing studies:

- Jourdan et al. (2006)
- Corne et al. (2012)
- Talbi (2016)
- Calvet et al. (2017)
- Song et al. (2019)
- Talbi (2022)

Main gaps:

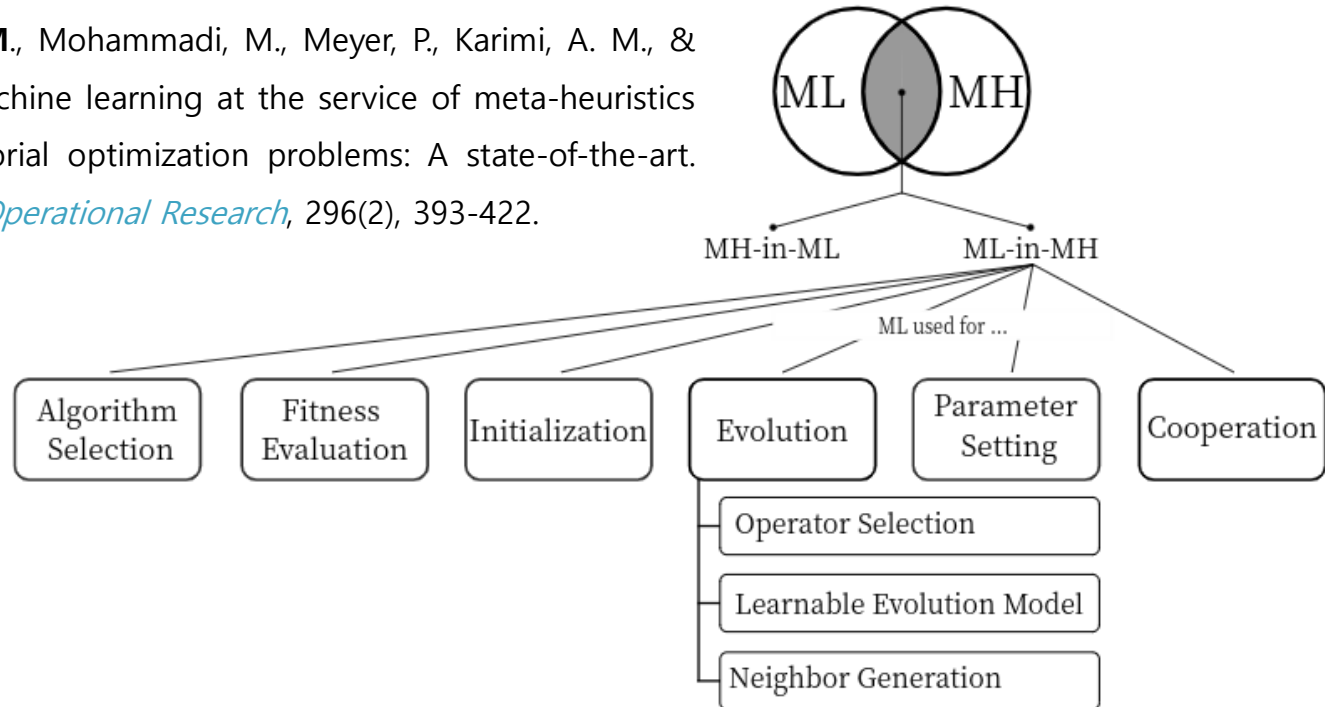
- No comprehensive taxonomy
- For only a specific purpose
- Lack of technical discussion

Q1. How and for which purposes machine learning can help meta-heuristics?

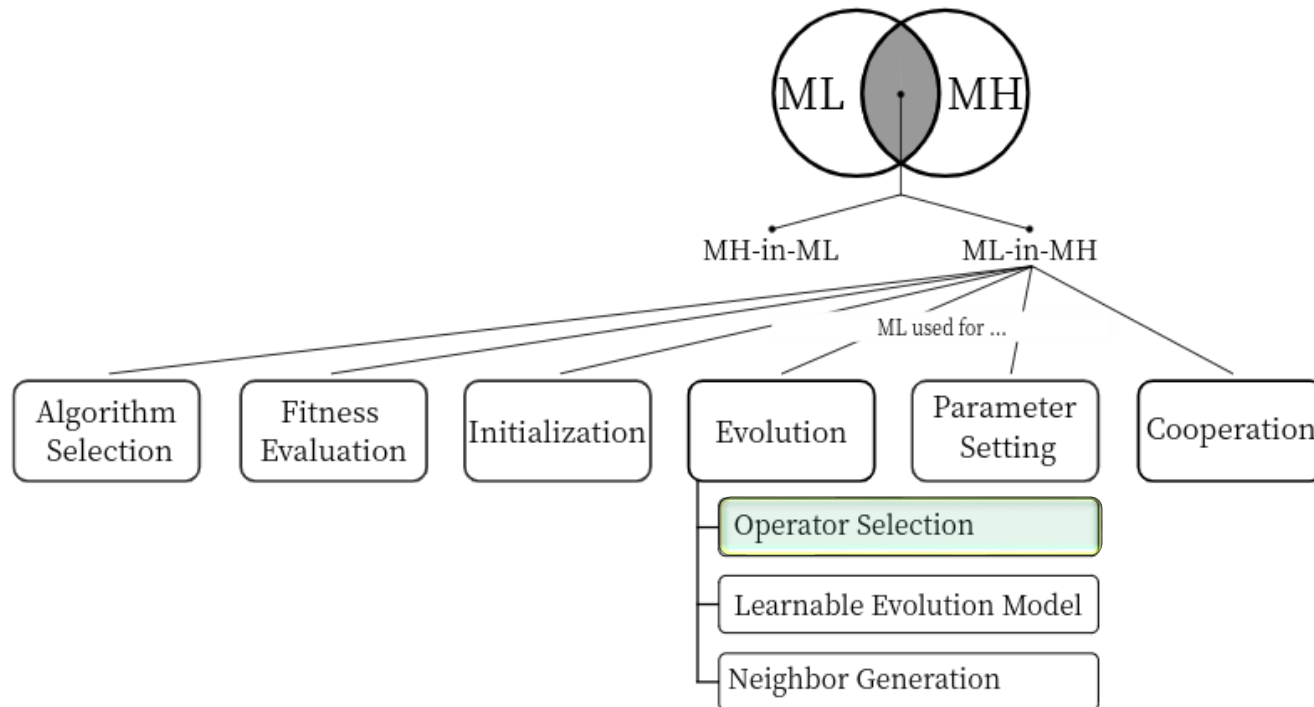


Q1. How and for which purposes machine learning can help meta-heuristics?

Karimi-Mamaghan, M., Mohammadi, M., Meyer, P., Karimi, A. M., & Talbi, E. G. (2022). Machine learning at the service of meta-heuristics for solving combinatorial optimization problems: A state-of-the-art. *European Journal of Operational Research*, 296(2), 393-422.



Q2. How machine learning helps **operator selection** in meta-heuristics?

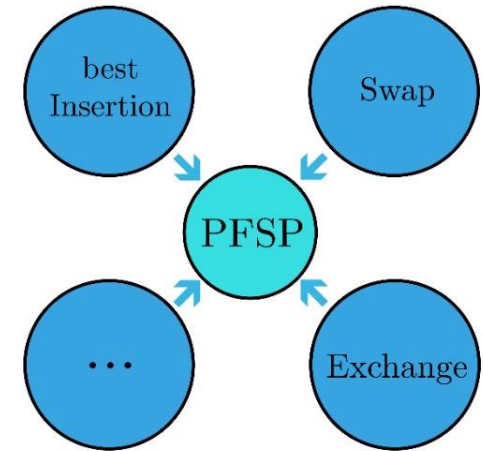


Q2. How machine learning helps **operator selection** in meta-heuristics?

- Multiple operators
- Non-stationary search space

Q2. How machine learning helps **operator selection** in meta-heuristics?

- Multiple operators
- Non-stationary search space



Q2. How machine learning helps **operator selection** in meta-heuristics?

- ▶ Random selection
- ▶ Sequential
- ▶ Adaptive operator selection (AOS)
 - Maturana et al. (2011)
 - Burke et al. (2011)
 - Da silva et al. (2016)
 - Ahmadi et al. (2018)

$$S_{t+1} = S_t + r_t$$

Q2. How machine learning helps **operator selection** in meta-heuristics?

- ▶ The success of a meta-heuristic depends on the choice of its operators
- ▶ Simple adaptive operator selection methods bring not much gain



Discrete Optimization

Meta-analysis of metaheuristics: Quantifying the effect of adaptiveness in adaptive large neighborhood search

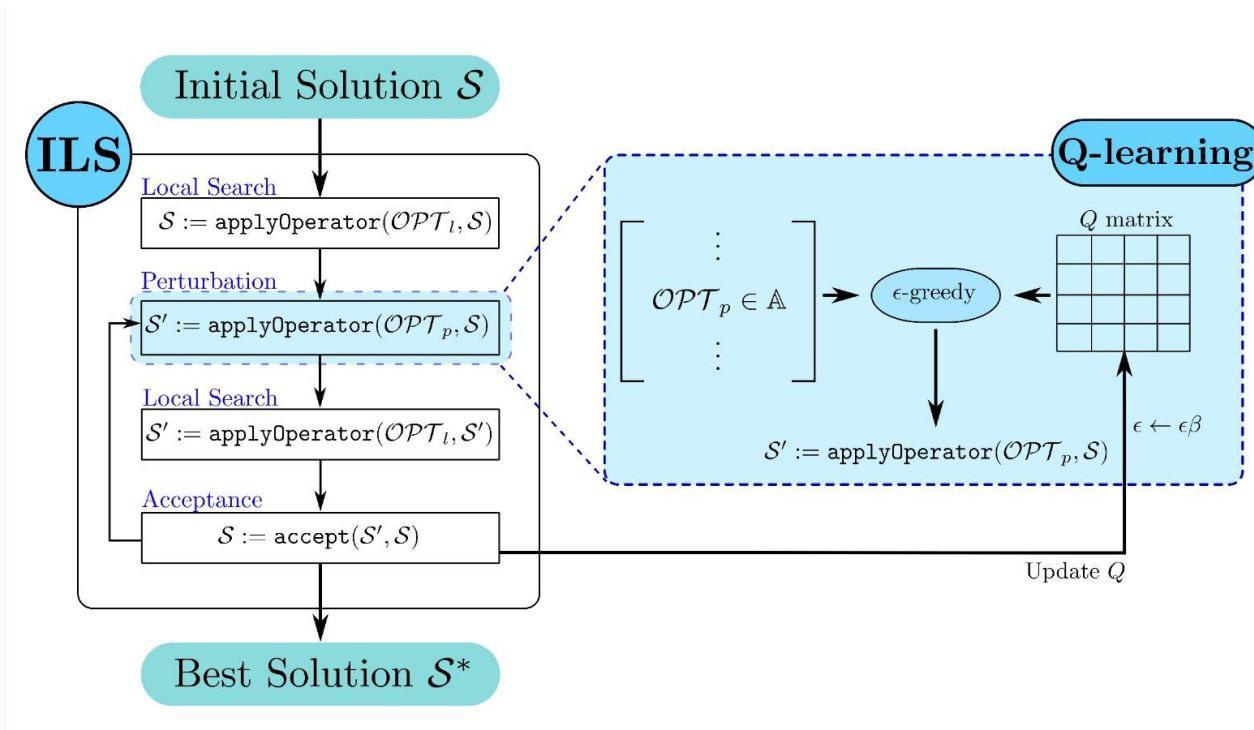
Renata Turkeš^{a,*}, Kenneth Sörensen^b, Lars Magnus Hvattum^c



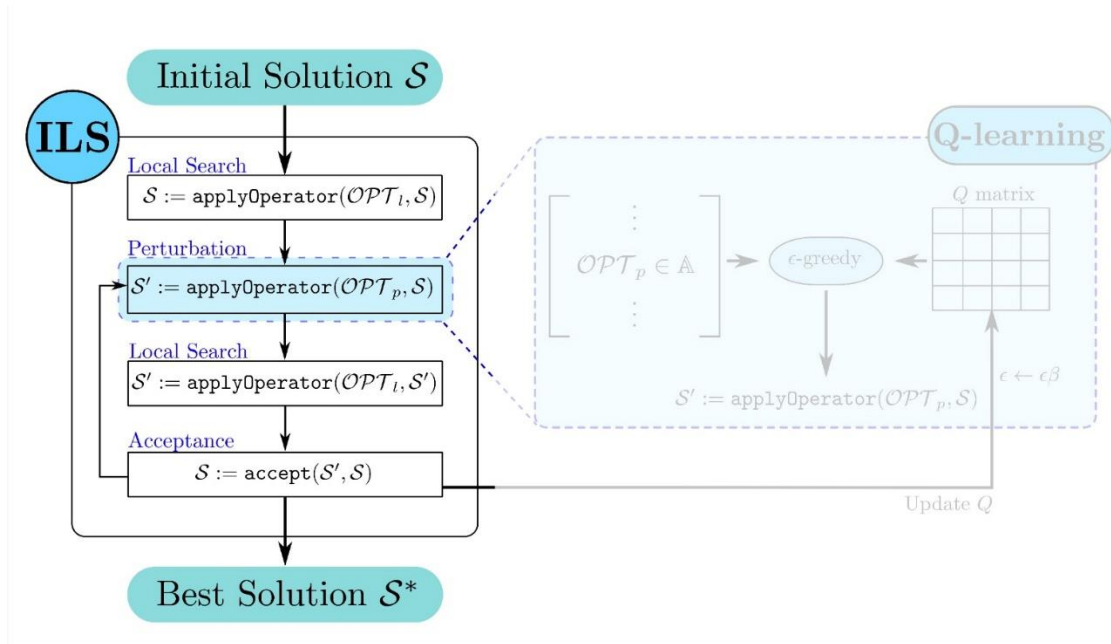
Learning to select operators

Proposed framework

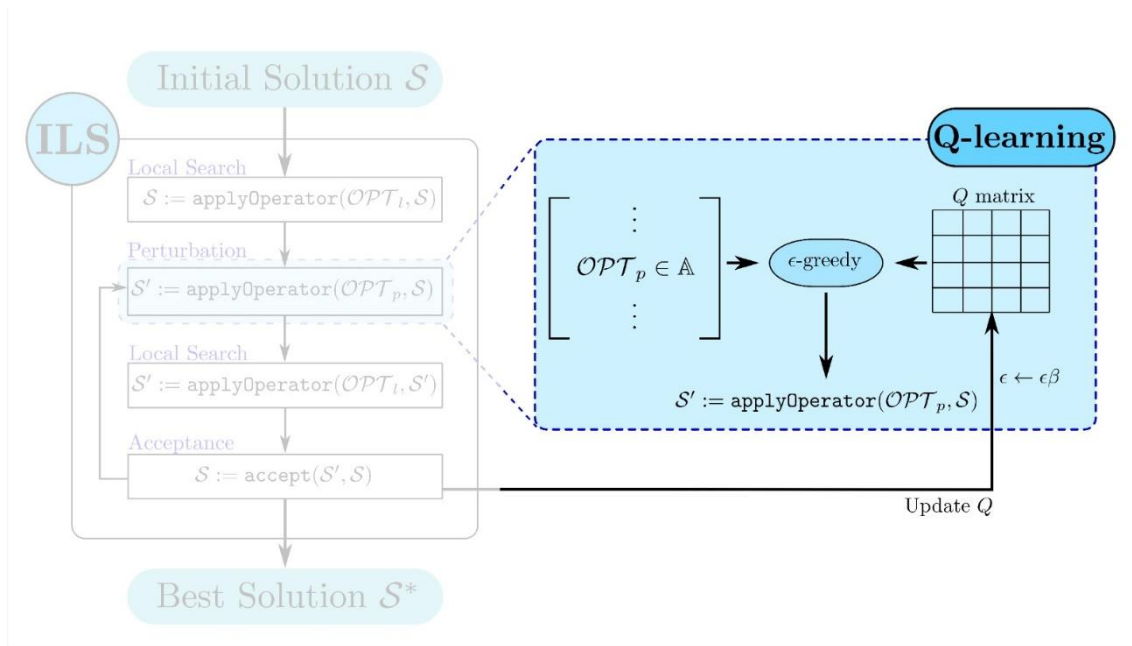
Integration of **Q-learning** into **Iterated Local Search** for operator selection



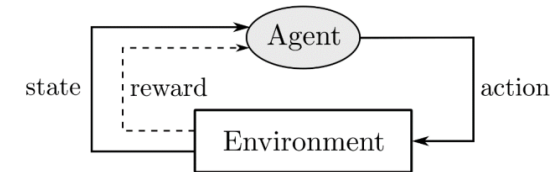
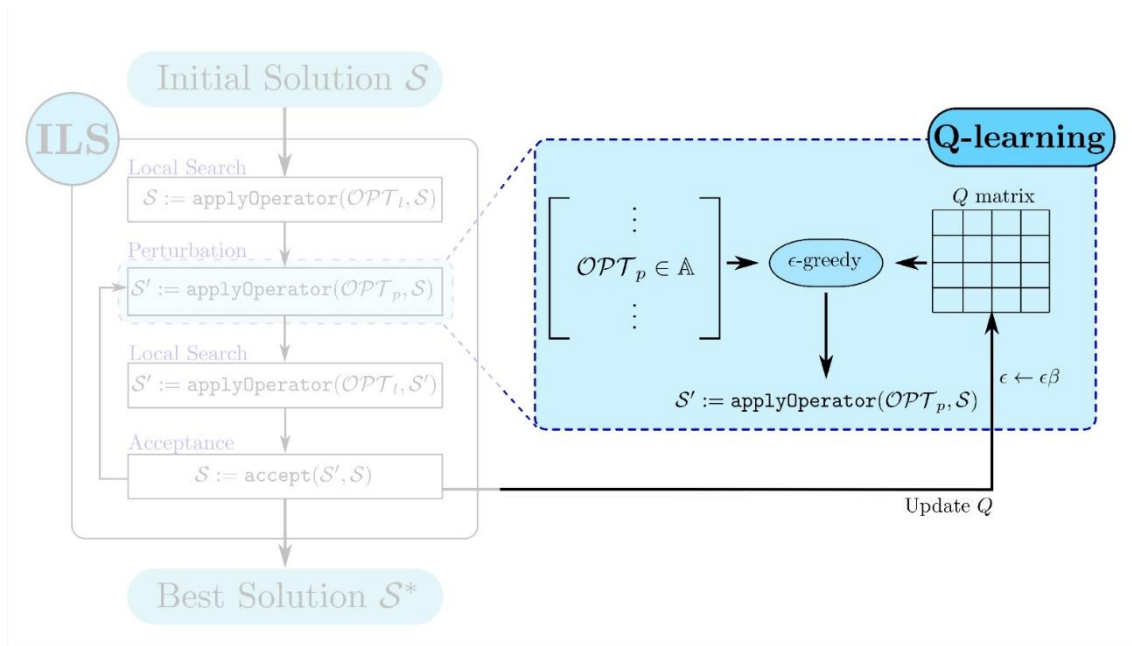
Integration of **Q-learning** into **Iterated Local Search** for operator selection



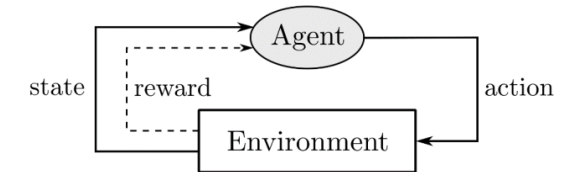
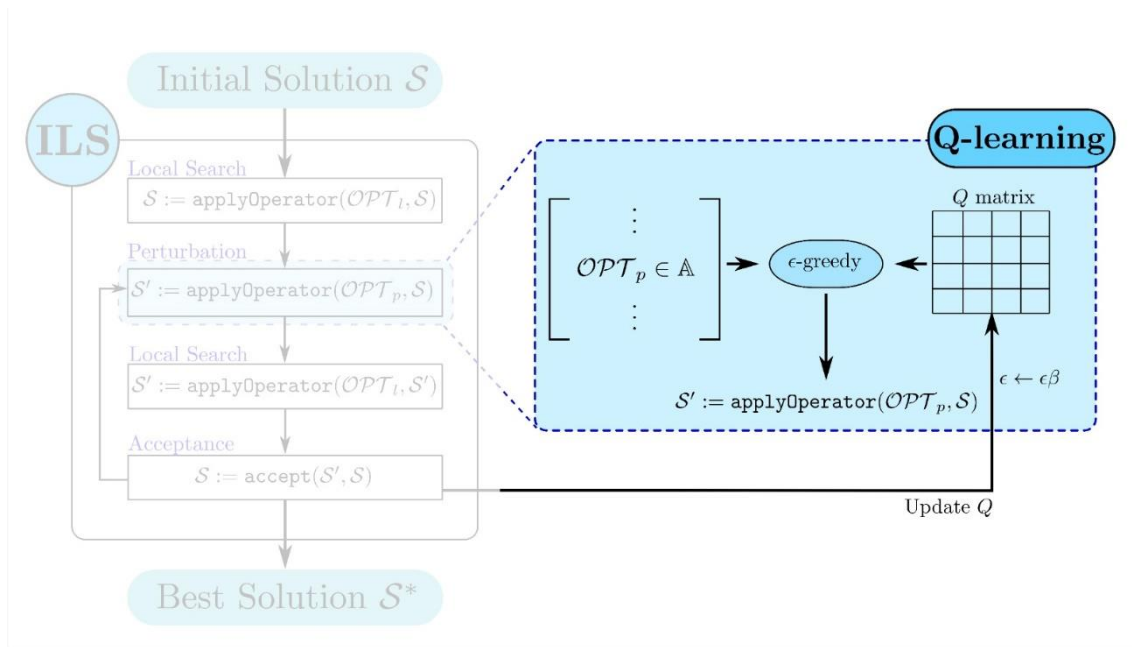
Integration of **Q-learning** into **Iterated Local Search** for operator selection



Integration of **Q-learning** into **Iterated Local Search** for operator selection

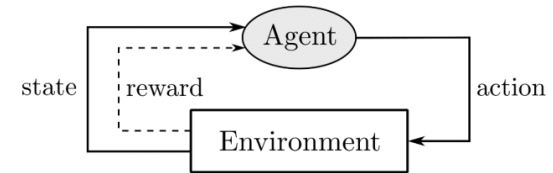
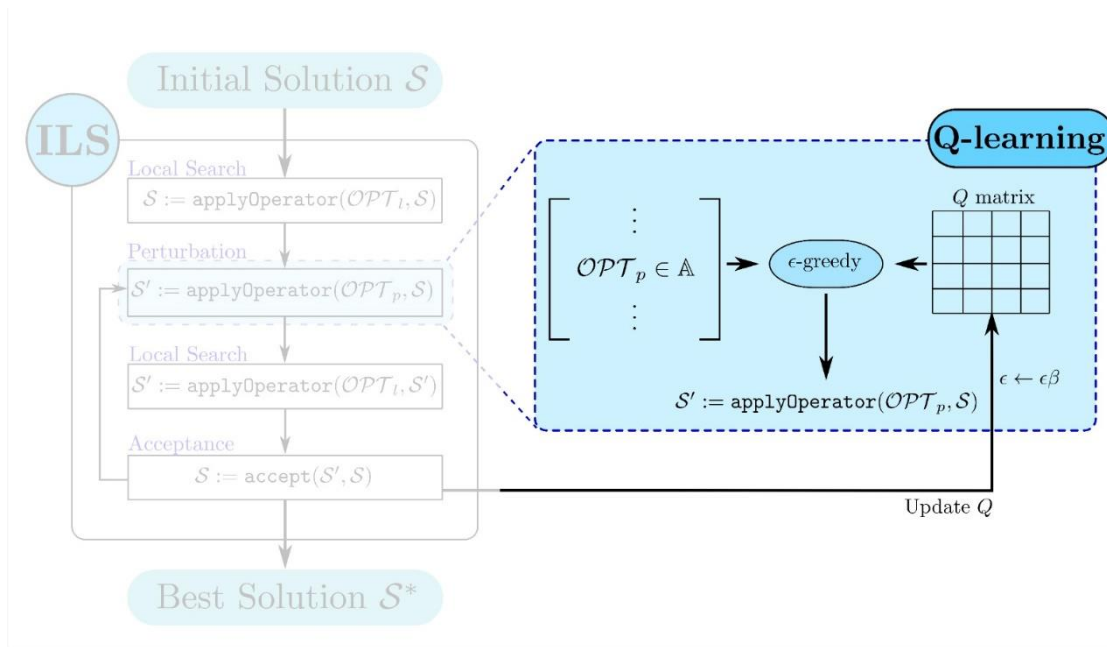


Integration of **Q-learning** into **Iterated Local Search** for operator selection

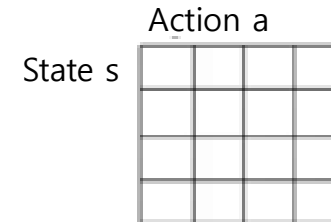


- Model-free

Integration of **Q-learning** into **Iterated Local Search** for operator selection

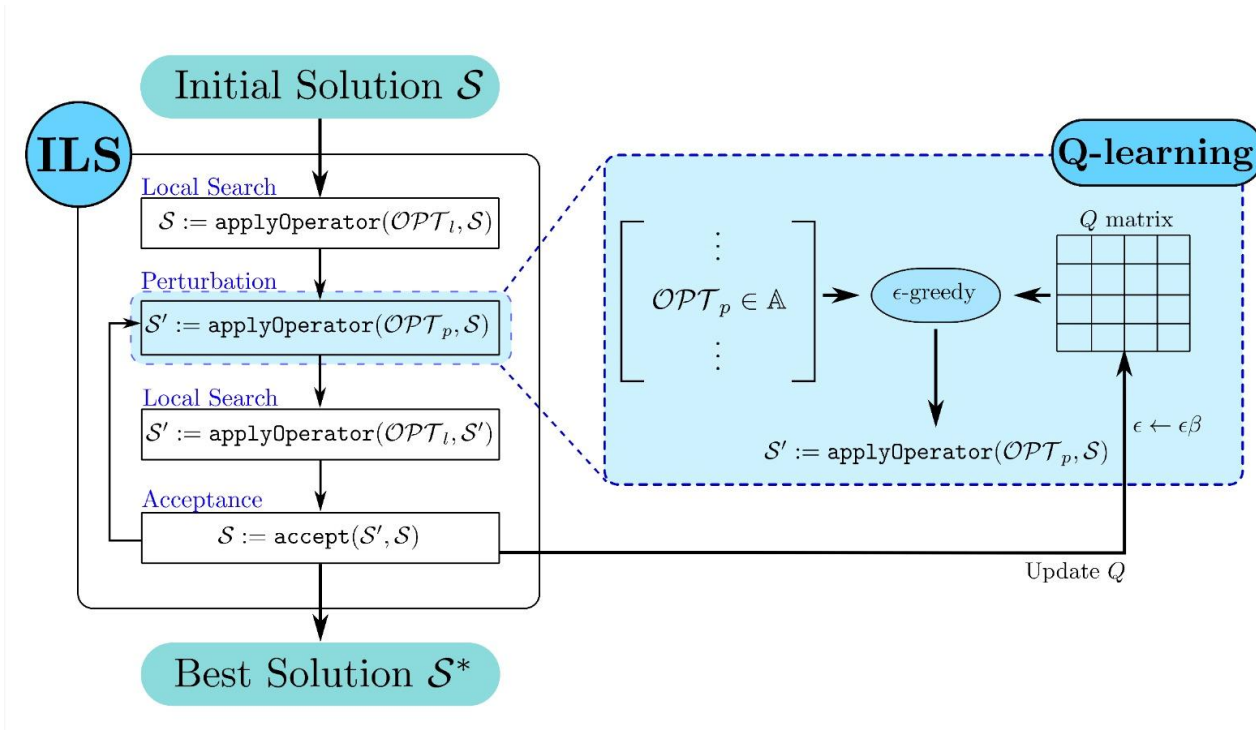


- Model-free
- Assigns and updates a Q-value

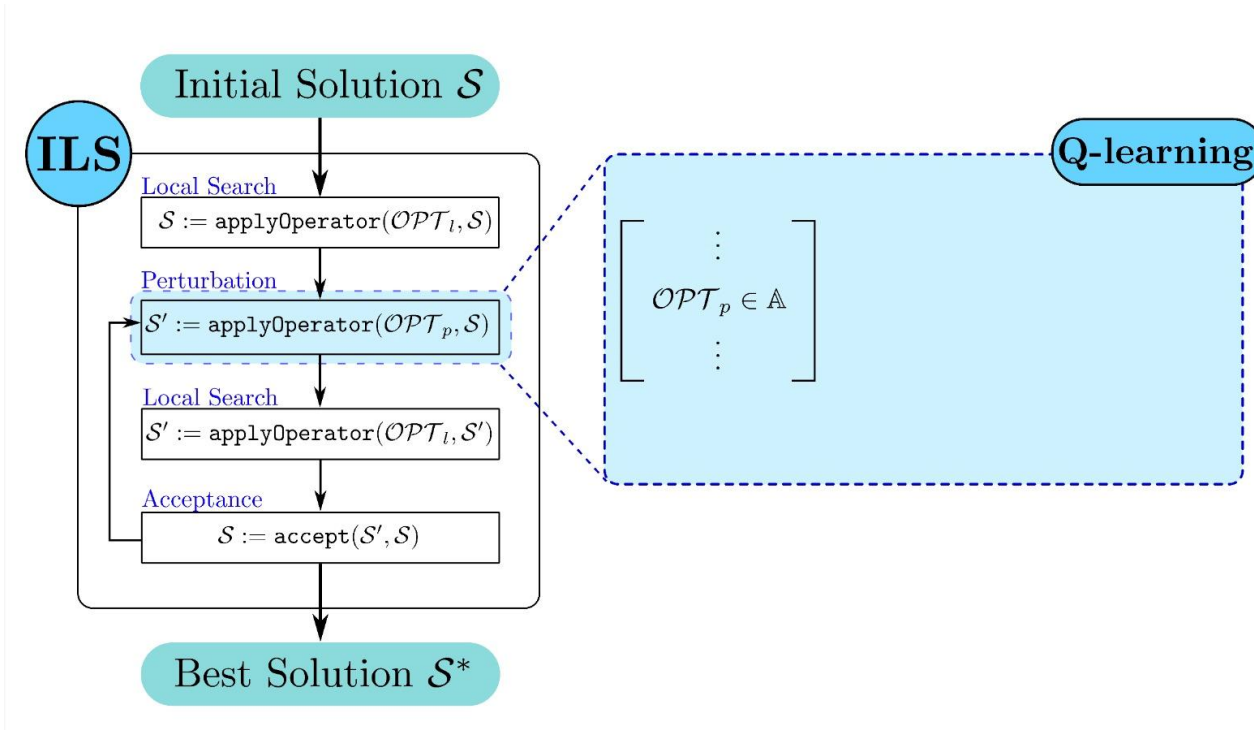


$$Q(s, a) = Q(s, a) + \alpha[r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a)]$$

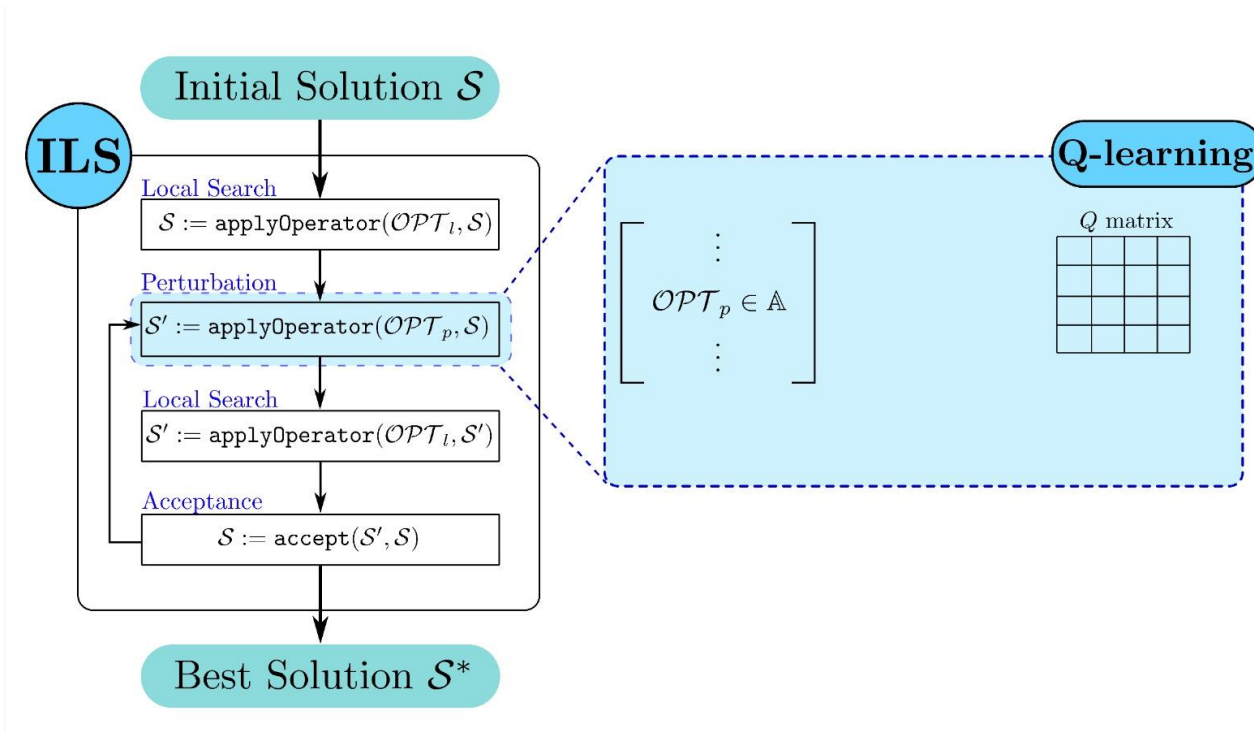
Integration of **Q-learning** into **Iterated Local Search** for operator selection (QILS)



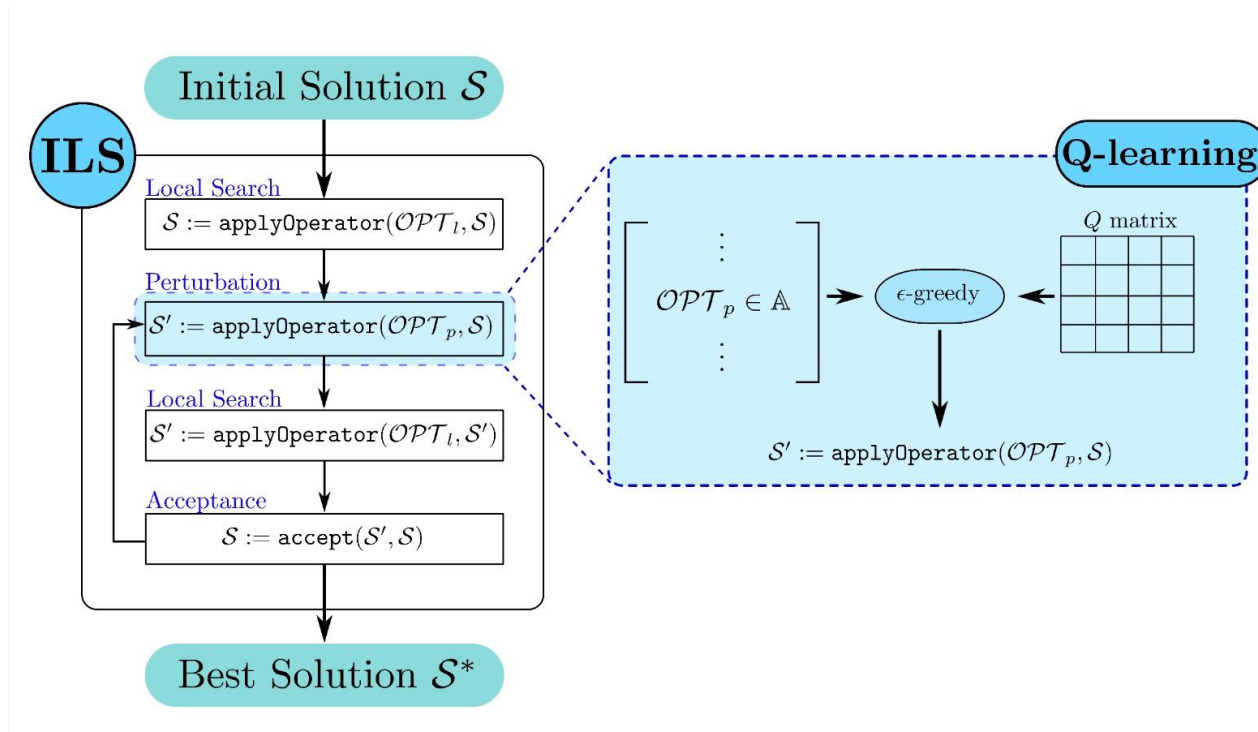
Integration of **Q-learning** into **Iterated Local Search** for operator selection (QILS)



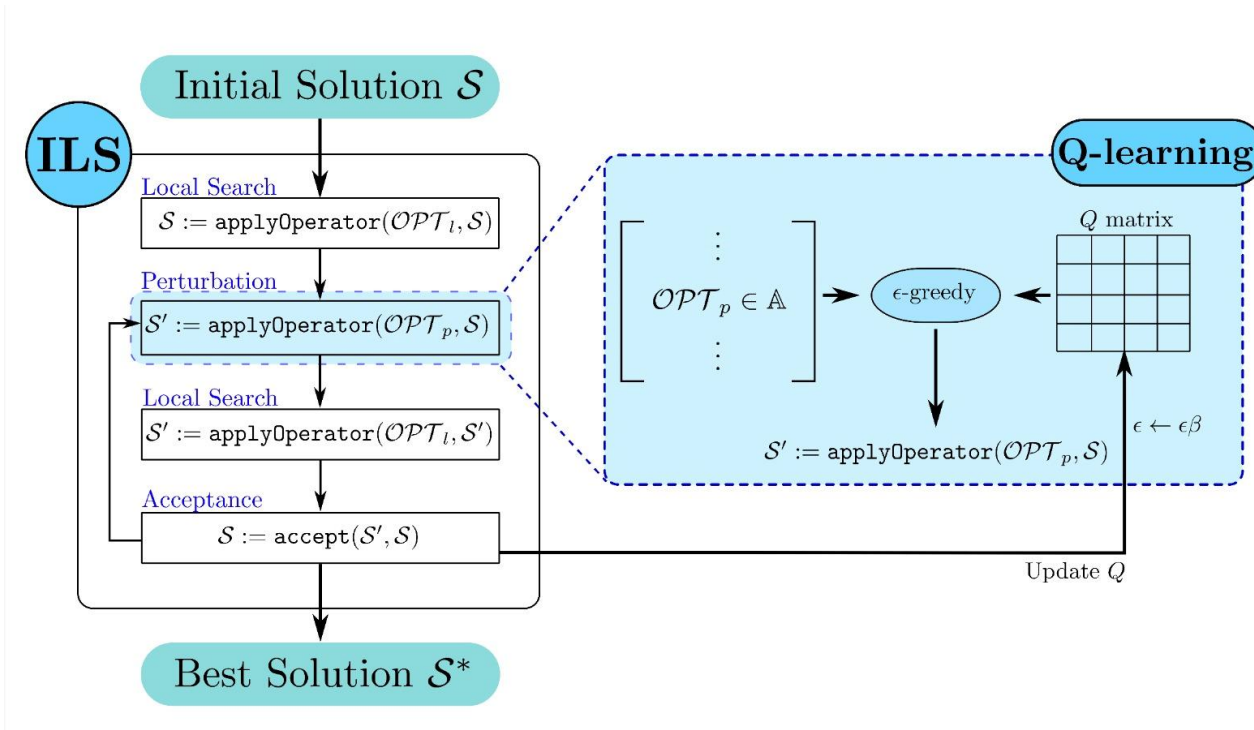
Integration of **Q-learning** into **Iterated Local Search** for operator selection (QILS)



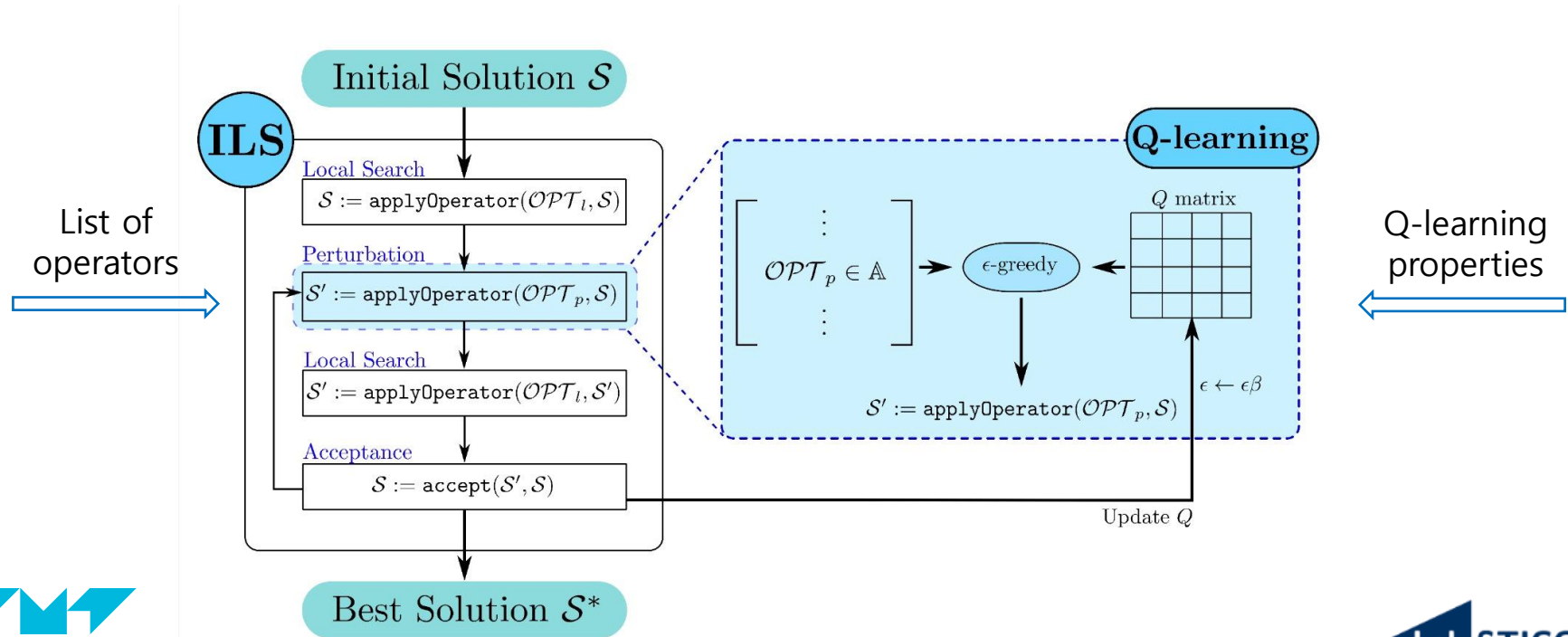
Integration of **Q-learning** into **Iterated Local Search** for operator selection (QILS)



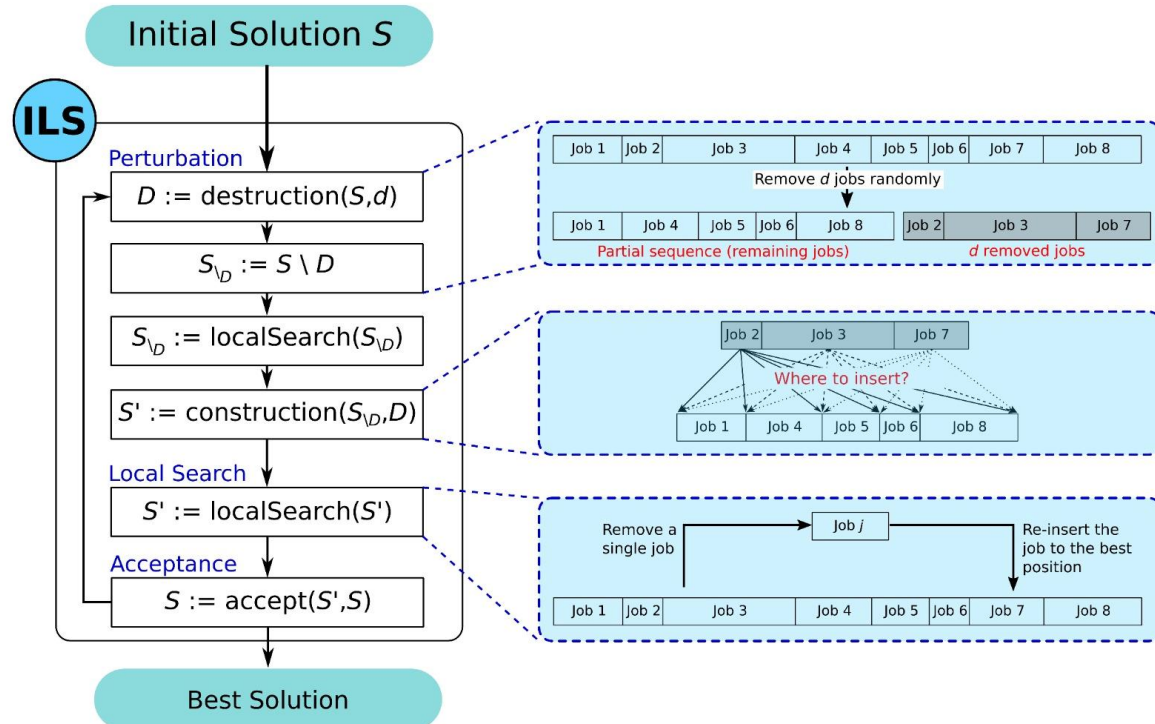
Integration of **Q-learning** into **Iterated Local Search** for operator selection (QILS)



Integration of **Q-learning** into **Iterated Local Search** for operator selection (QILS)



- Perturbation operator
Destruction-construction with multiple values of d
- Local search operator:
Best-insertion

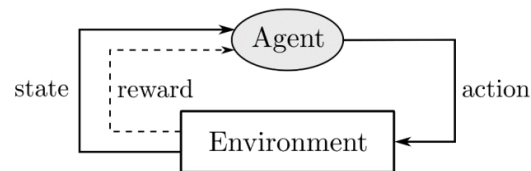


Action

- $A = \{1, 2, \dots, d_{max}\}$
- A set of different perturbation operators with different strength

State

- $S = \{0, 1\}$
- $S = 1$, if the perturbation operator brings out the solution from the local optimum
- $S = 0$, otherwise



Reward

- **Local** and **Global** reward (improvement in objective function value)

$$r = \omega \cdot r_{local} + (1 - \omega) r_{global}$$

Action selection

- ϵ - greedy strategy

$$a = \begin{cases} \operatorname{argmax}_{a \in A} Q(s, a) & \text{with probability } 1 - \epsilon \\ \text{any action selected uniformly and randomly in } A & \text{with probability } \epsilon \end{cases}$$

Learning to select operators

Experimental design

Q2. How machine learning helps **operator selection** in meta-heuristics?

1. Effectiveness of multiple operators
2. Effectiveness of Q-learning based operator selection
3. Competitiveness in comparison to the state-of-the-art
4. Adaptiveness to the search space

Phase 1 (QILS vs its non-learning versions): Stopping criterion: Fixed number of iteration

1. Effectiveness of multiple perturbation operators:

ILS with Random operator selection (RILS) vs. ILS with Individual operator (IILS)

2. Effectiveness of Q-learning based operator selection:

QILS vs. RILS

Phase 1 (QILS vs its non-learning versions): Stopping criterion: Fixed number of iteration

Phase 2 (QILS vs literature): Stopping criterion: Fixed CPU Time

3. Competitiveness of QILS

QILS vs. state-of-the-art

4. Adaptiveness of QILS

Benchmark datasets:

- **Taillard:** 120 instances (**20** jobs, **5** machines) to (**500** jobs, **20** machines)
- **VRF-hard-small:** 240 instances (**10** jobs, **5** machines) to (**60** jobs, **20** machines)
- **VRF-hard-large:** 240 instances (**100** jobs, **20** machines) to (**800** jobs, **60** machines)

Benchmark datasets:

- **Taillard:** 120 instances (**20** jobs, **5** machines) to (**500** jobs, **20** machines)
- **VRF-hard-small:** 240 instances (**10** jobs, **5** machines) to (**60** jobs, **20** machines)
- **VRF-hard-large:** 240 instances (**100** jobs, **20** machines) to (**800** jobs, **60** machines)

Performance metrics:

- Average Relative Percentage Deviation (ARPD)
- Worst-case complexity
- Computational time
- Convergence behavior

$$RPD = \frac{OF - OF^*}{OF^*} \times 100$$

Benchmark datasets:

- **Taillard:** 120 instances (**20** jobs, **5** machines) to (**500** jobs, **20** machines)
- **VRF-hard-small:** 240 instances (**10** jobs, **5** machines) to (**60** jobs, **20** machines)
- **VRF-hard-large:** 240 instances (**100** jobs, **20** machines) to (**800** jobs, **60** machines)

Performance metrics:

- Average Relative Percentage Deviation (ARPD)
- Worst-case complexity
- Computational time
- Convergence behavior

$$RPD = \frac{OF - OF^*}{OF^*} \times 100$$

Statistical test (Wilcoxon signed rank test with 95% confidence level)

- Non-parametric test & Pairwise comparisons

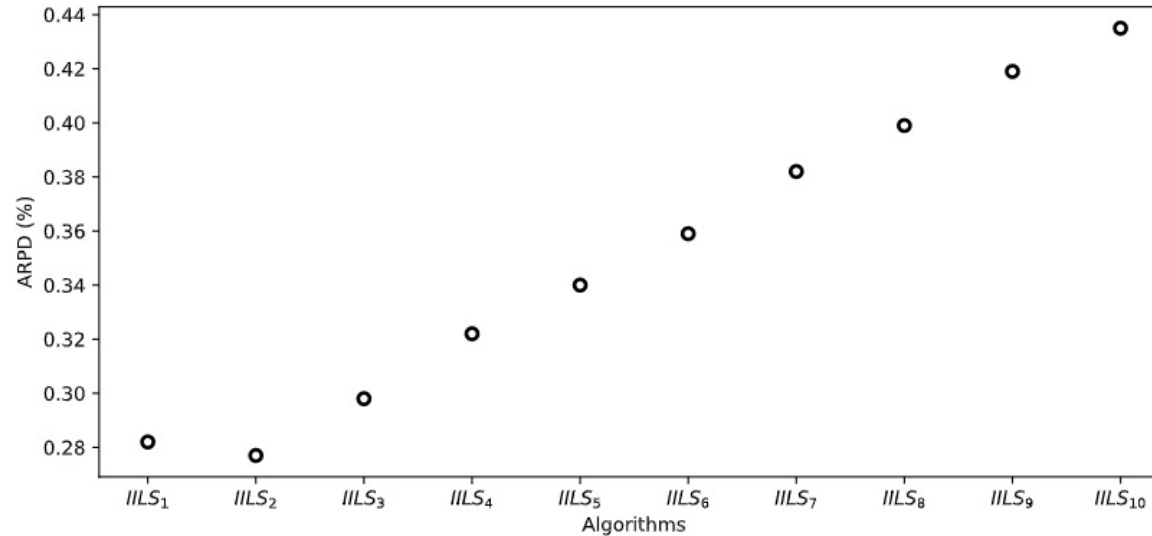
Parameter tuning:

- Algorithm's parameters
 - ϵ (Epsilon)
 - β (Epsilon decay)
 - α (Learning rate)
 - γ (Discount factor)
 - E (size of episode)
 - w (local/global reward weight)
 - τ (acceptance parameter)
- Design of experiment: 3k factorial Box–Behnken design (62 experiments for 7 variable design)
- RSM - Response Surface Methodology

Parameter tuning:

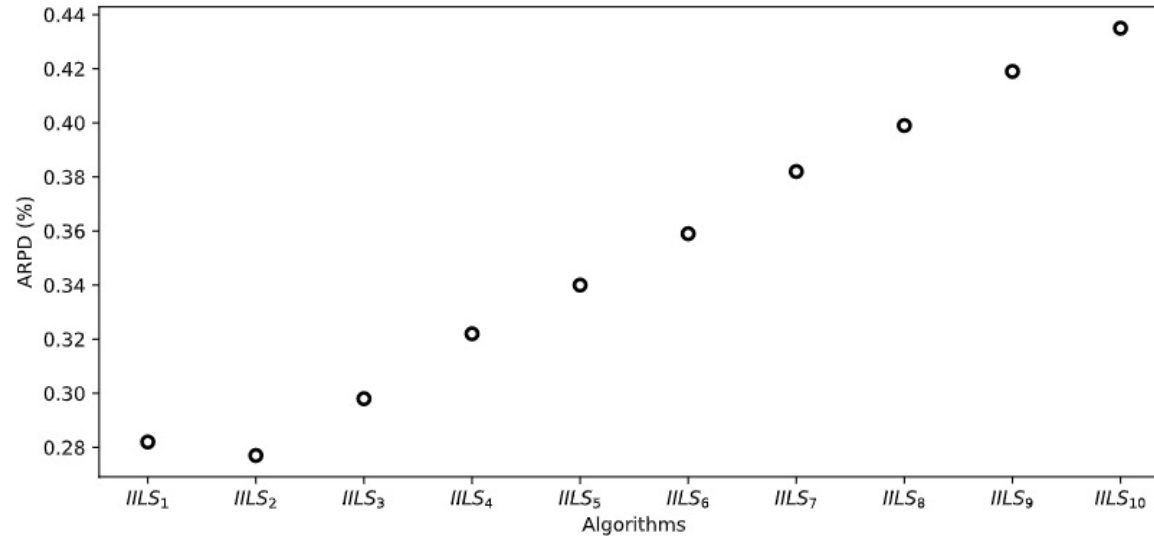
- Problem's parameter
 - d_{max} (maximum jobs to remove)

$$A = \{1, 2, \dots, d_{max}\}$$



Parameter tuning:

- Problem's parameter
 - d_{max} (maximum jobs to remove) = 3



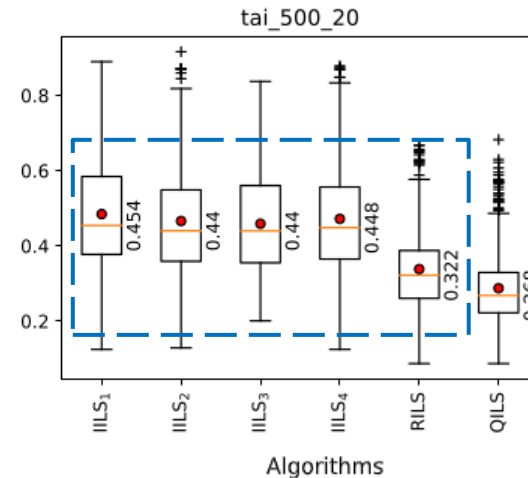
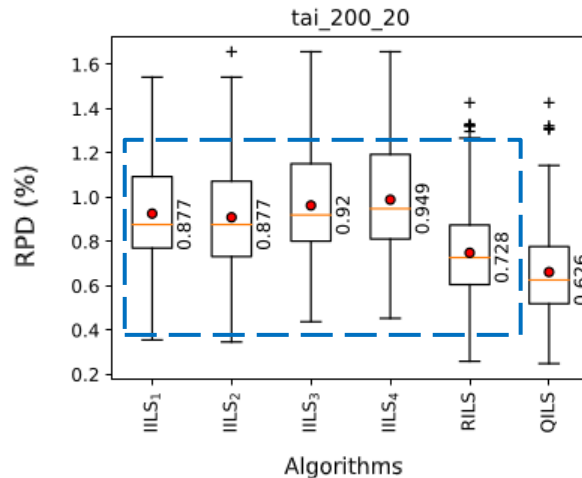
Learning to select operators

Numerical results

Phase 1 (QILS vs its non-learning versions):

1. Effectiveness of multiple perturbation operators: **RILS** vs. **IILS**
2. Effectiveness of Q-learning based operator selection: **QILS** vs. **RILS**

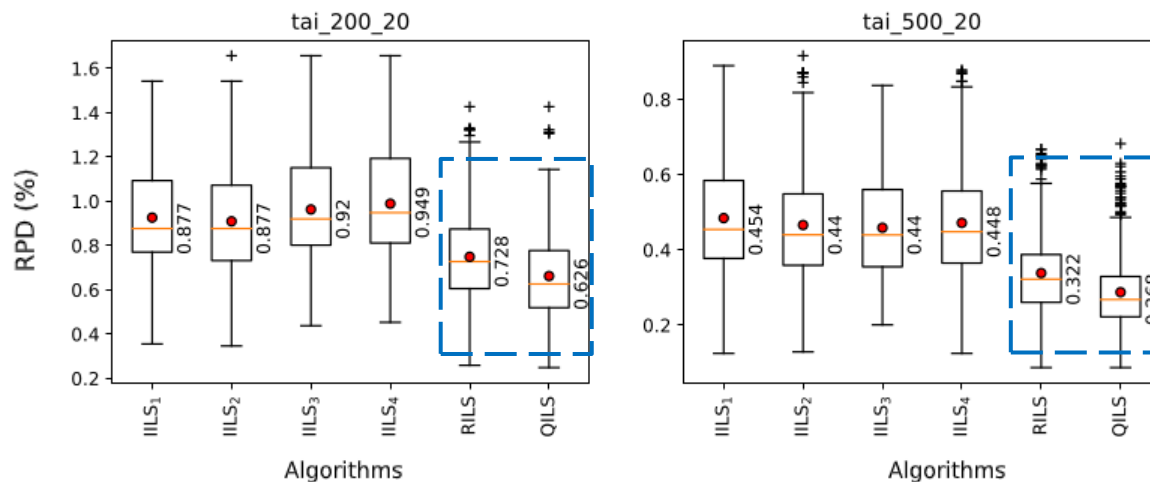
Average Relative Percentage Deviation (ARPD)



Phase 1 (QILS vs its non-learning versions):

1. Effectiveness of multiple perturbation operators: **RILS** vs. **IILS**
2. Effectiveness of Q-learning based operator selection: **QILS** vs. **RILS**

Average Relative Percentage Deviation (ARPD)



Phase 1 (QILS vs its non-learning versions):

1. Effectiveness of multiple perturbation operators: **RILS** vs. **IILS**
2. Effectiveness of Q-learning based operator selection: **QILS** vs. **RILS**

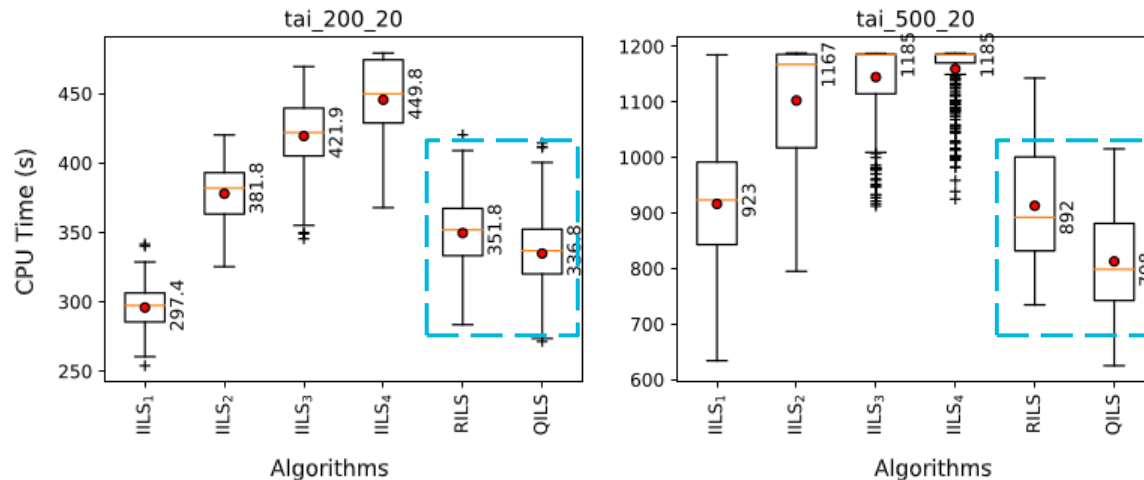
Worst-case complexity

Q-learning overhead: $O(Itr_{max} \times d_{max})$

Phase 1 (QILS vs its non-learning versions):

1. Effectiveness of multiple perturbation operators: **RILS** vs. **IILS**
2. Effectiveness of Q-learning based operator selection: **QILS** vs. **RILS**

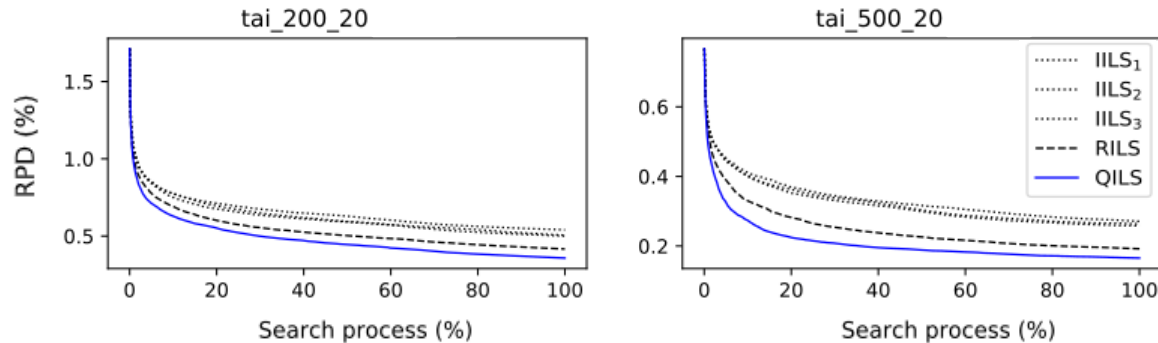
Computational time



Phase 1 (QILS vs its non-learning versions):

1. Effectiveness of multiple perturbation operators: **RILS** vs. **IILS**
2. Effectiveness of Q-learning based operator selection: **QILS** vs. **RILS**

Convergence



Phase 1 (QILS vs its non-learning versions):

1. Effectiveness of multiple perturbation operators: **RILS** vs. **IILS**
2. Effectiveness of Q-learning based operator selection: **QILS** vs. **RILS**

Overall comparison

Dataset	Algorithms					
	IILS ₁	IILS ₂	IILS ₃	IILS ₄	RILS	QILS
Taillard	0.285	0.281	0.302	0.325	0.246	0.223
VRFB-hard-small	0.167	0.154	0.172	0.202	0.146	0.133
VRFB-hard-large	0.109	0.125	0.165	0.212	-0.001	-0.073
Average	0.187	0.187	0.213	0.246	0.130	0.094
Improvement(%)	49	50	55	61	28	

Phase 2 (QILS vs literature):

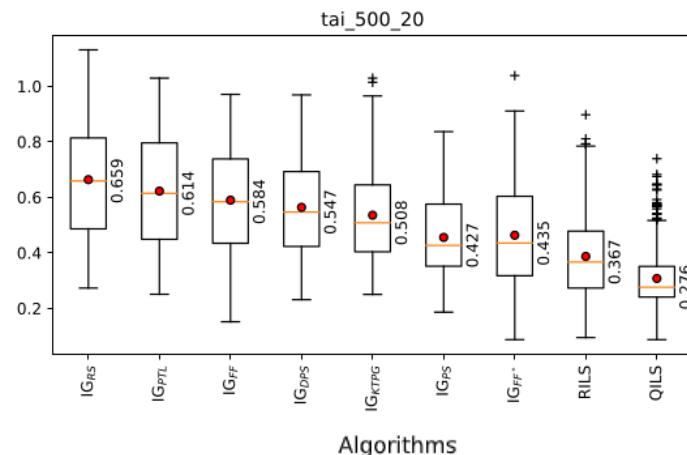
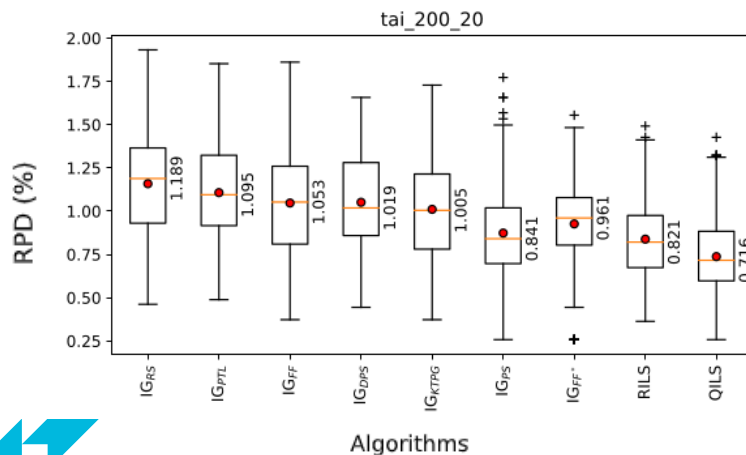
3. Competitiveness of QILS: **QILS** vs. **state-of-the-art**

- **IG_{FF*}** (Viagas and Framinan 2019)
- **IG_{PS}** (Pagnozzi & Stützle 2019)
- **IG_{KTPG}** (Kizilay et al. 2019)
- **IG_{DPS}** (Dubois et al. 2017)
- **IG_{FF}** (Viagas and Framinan 2014)
- **IG_{PTL}** (Pan et al. 2008)
- **IG_{RS}** (Ruiz & Stutzle 2007)

Phase 2 (QILS vs literature):

3. Competitiveness of QILS: **QILS** vs. **state-of-the-art**

Average Relative Percentage Deviation (ARPD)



Phase 2 (QILS vs literature):

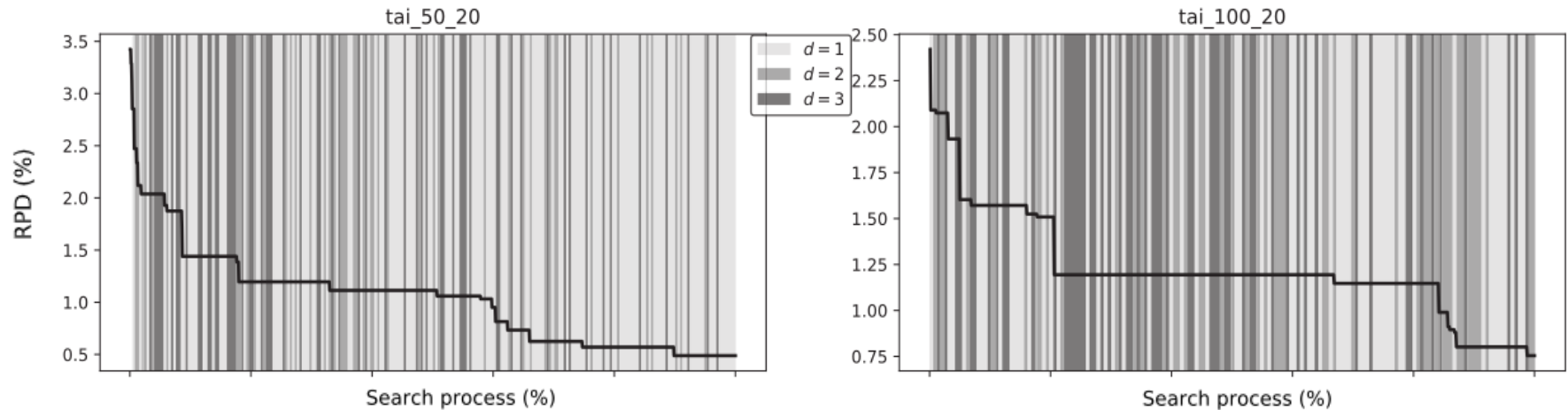
3. Competitiveness of QILS: **QILS** vs. **state-of-the-art**

Overall comparison

Dataset	Algorithm								
	IG _{RS}	IG _{PTL}	IG _{FF}	IG _{DPS}	IG _{KTPG}	IG _{PS}	IG _{FF*}	RILS	QILS
Taillard	0.322	0.319	0.304	0.302	0.285	0.267	0.271	0.256	0.228
VRP-small	0.151	0.161	0.148	0.127	0.126	0.122	0.123	0.122	0.110
VRP-large	0.509	0.623	0.481	0.200	0.131	0.101	0.109	0.074	-0.013
Average	0.327	0.368	0.311	0.210	0.181	0.163	0.168	0.151	0.108
Improvement(%)	66	70	65	48	40	33	35	29	

4. Adaptiveness of QILS

- Different operators at different stages



- **Karimi-Mamaghan, M.**, Mohammadi, M., Meyer, P., Karimi-Mamaghan, A. M., & Talbi, E. G. (2022). Machine learning at the service of meta-heuristics for solving combinatorial optimization problems: A state-of-the-art. *European Journal of Operational Research*, 296(2), 393-422.
- **Karimi-Mamaghan, M.**, Mohammadi, M., Meyer, P., & Passet, B. Learning to select operators in meta-heuristics: An integration of Q-learning into the iterated greedy algorithm for the permutation flowshop scheduling problem. *European Journal of Operational Research*, 2022.
- **Karimi-Mamaghan, M.**, Passet, B., Mohammadi, M., & Meyer, P. A Learning-based Iterated Local Search Algorithm for Solving the Traveling Salesman Problem. In *OLA 2021 International Conference on Optimization and learning, Springer volume of Communications in Computer and Information Science (CCIS) series*, 2021.
- **Karimi-Mamaghan, M.**, Mohammadi, M., Jula, P., Pirayesh, A., & Ahmadi, H. A Learning-based Metaheuristic for a Multi-objective Agile Inspection Planning Model under Uncertainty. *European Journal of Operational Research*, 285(2), 513-537, 2020.
- **Karimi-Mamaghan, M.**, Mohammadi, M., Pirayesh, A., Karimi-Mamaghan, A. M., & Irani, H. Hub-and-spoke network design under congestion: A learning based metaheuristic. *Transportation Research Part E: Logistics and Transportation Review*, 142, 102069, 2020.

- **Karimi-Mamaghan, M.**, Mohammadi, M., Pasdeloup, B., & Meyer, P. Q-learning for operator selection in meta-heuristics: an application to the permutation flowshop scheduling problem . In [Canadian Operational Research Society Annual Conference \(CORS 2022\)](#), Jun 2022, Vancouver, Canada (Oral presentation).
- **Karimi-Mamaghan, M.**, Meyer, P., Mohammadi, M., & Pasdeloup, B. A reinforcement learning-based operator selection in iterated local search for solving scheduling problems. In [31st European Conference on Operational Research \(EURO 2021\)](#), Jul 2021, Athens, Greece (Oral presentation).
- **Karimi-Mamaghan, M.**, Pasdeloup, B., Mohammadi, M., & Meyer, P. Learning-based Iterated Local Search Algorithm for Solving the Traveling Salesman Problem. In [OLA 2021 International Conference on Optimization and learning](#) (online), 2021.
- **Karimi-Mamaghan, M.**, Mohammadi, M., Pasdeloup, B., Billot, R., & Meyer, P. An Online Learning-based Metaheuristic for Solving Combinatorial Optimization Problems. In [ROADEF 2020: 21ème Congrès Annuel de la Société Française de Recherche Opérationnelle et d'Aide à la Décision](#), 2020, February, (Oral presentation).

Learning to select operators

Discussion and conclusion

- ▶ Automatic selection without the need for expert knowledge
- ▶ Especially useful when dealing with competitive operators
- ▶ No need for training and no extra computational effort
- ▶ Adapt the behavior of meta-heuristic to the problem instance
- ▶ Simple to use and easy to adapt to solve any problem



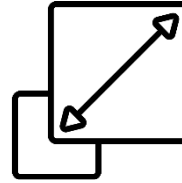
Data availability



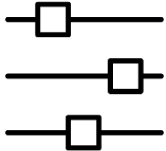
Extra complexity overhead



Processing of massive data



Scaling to larger problem



Extra parameters

- ▶ A set of additional parameters of ML (A one-time task)
- ▶ How to define the set of states and reward? (The goal of using Q-learning)
- ▶ When numerous operators are available (we are currently working to adapt the framework to a **numerous operators**)
- ▶ Only applied to two problems (we are currently working on the idea of **generalizability**)

Learning to select operators

Future research

Generalizability of the proposed framework

- ▶ Hyper-heuristic: a high-level automated search methodology that automates design/selection of heuristics
- ▶ Perform search over the space of a set of heuristics, instead of the solution space
- ▶ Implemented the framework in HyFlex, java interface for hyper-heuristics (Ochoa et al. 2012)

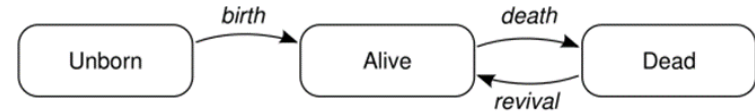
Research visit at the University of Nottingham (UK), Computational Optimization and Learning lab

Prof. Ender Ozcan

When numerous operators are available

► Dynamic operator management

- Consider a dynamic set of operators instead of a static one
- Inefficient are disabled, new operators are added
- Different states for operators: unborn, alive, dead



Research visit at Polytechnique Montreal (Canada), Department of Mathematical and Industrial Engineering

Prof. Nadia Lahrichi & Prof. Andrea Lodi

Thank you for your attention

- Ahmadi, E., Goldengorin, B., Süer, G. A., & Mosadegh, H. (2018). A hybrid method of 2-tsp and novel learning-based ga for job sequencing and tool switching problem. *Applied soft computing*, 65 , 214–229 .
- Burke, E. K. , Gendreau, M. , Ochoa, G. , & Walker, J. D. (2011). Adaptive iterated local search for cross-domain optimisation. In *Proceedings of the 13th annual conference on genetic and evolutionary computation* (pp. 1987–1994) .
- Da Silva, A. M. L. , Freire, M. R. , & Honório, L. M. (2016). Transmission expansion planning optimization by adaptive multi-operator evolutionary algorithms. *Electric Power Systems Research*, 133 , 173–181.
- Dubois-Lacoste, J., Pagnozzi, F., & Stützle, T. (2017). An iterated greedy algorithm with optimization of partial solutions for the makespan permutation flowshop problem. *Computers & Operations Research*, 81, 160-166.
- Fernandez-Viagas, V., & Framinan, J. M. (2014). On insertion tie-breaking rules in heuristics for the permutation flowshop scheduling problem. *Computers & Operations Research*, 45, 60-67.
- Fernandez-Viagas, V., & Framinan, J. M. (2019). A best-of-breed iterated greedy for the permutation flowshop scheduling problem with makespan objective. *Computers & Operations Research*, 112 , 104767 .
- Kizilay, D., Tasgetiren, M. F., Pan, Q.-K., & Gao, L. (2019). A variable block insertion heuristic for solving permutation flow shop scheduling problem with makespan criterion. *Algorithms*, 12 (5), 100 .
- Maturana, J. , Fialho, A. , Saubion, F. , Schoenauer, M. , Lardeux, F. , & Sebag, M. (2011). Adaptive operator selection and management in evolutionary algorithms. In *Autonomous search* (pp. 161–189). Springer .
- Pagnozzi, F., & Stützle, T. (2019). Automatic design of hybrid stochastic local search algorithms for permutation flowshop problems. *European journal of operational research*, 276 (2), 409–421.
- Pan, Q. K., Tasgetiren, M. F., & Liang, Y. C. (2008). A discrete differential evolution algorithm for the permutation flowshop scheduling problem. *Computers & Industrial Engineering*, 55(4), 795-816.
- Ruiz, R., & Stützle, T. (2007). A simple and effective iterated greedy algorithm for the permutation flowshop scheduling problem. *European journal of operational research*, 177(3), 2033-2049.



IMT Atlantique

Bretagne-Pays de la Loire
École Mines-Télécom



Hybridizing Metaheuristics with Machine Learning for Combinatorial Optimization: A taxonomy and Learning to Select Operators

Maryam KARIMI MAMAGHAN

To obtain the PhD degree

Reviewers:

Jin-Kao HAO
Kenneth SORRENSEN

Jury members: Olga BATTIA

Axel PARMENTIER
Sébastien VEREL

Director:

Patrick MEYER

Supervisors:

Mehrdad MOHAMMADI
Bastien PASDELOUP

July 20, Brest, France

Action

- $A = \{\text{Local search operators}\}$
- A set of different local search operators

State

- S = sequence of last k selected local search operator ($k=1$)

Reward

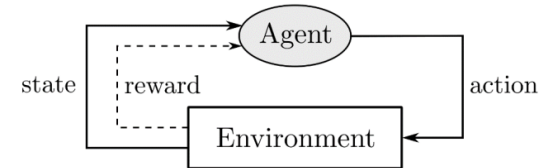
- **Local** and **Global** reward (improvement in objective function value)

$$r = \omega \cdot r_{local} + (1 - \omega) r_{global}$$

Action selection

- ϵ - greedy strategy

$$a = \begin{cases} \operatorname{argmax}_{a \in A} Q(s, a) & \text{with probability } 1 - \epsilon \\ \text{any action selected uniformly and randomly in } A & \text{with probability } \epsilon \end{cases}$$

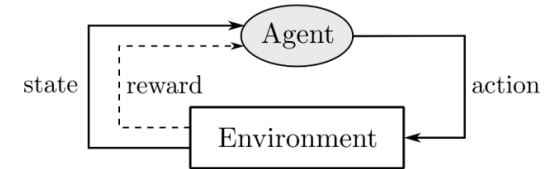


Action

- $A = \{\text{perturbation operators}\}$
- A set of different perturbation search operators

State

- $S = \{0, 1\}$
- $S = 1$, if the perturbation operator brings out the solution from the local optimum
- $S = 0$, otherwise



Reward

- **Local** and **Global** reward (improvement in objective function value)

$$r = \omega \cdot r_{local} + (1 - \omega) r_{global}$$

Action selection

- ϵ – greedy strategy

$$a = \begin{cases} \operatorname{argmax}_{a \in A} Q(s, a) & \text{with probability } 1 - \epsilon \\ \text{any action selected uniformly and randomly in } A & \text{with probability } \epsilon \end{cases}$$

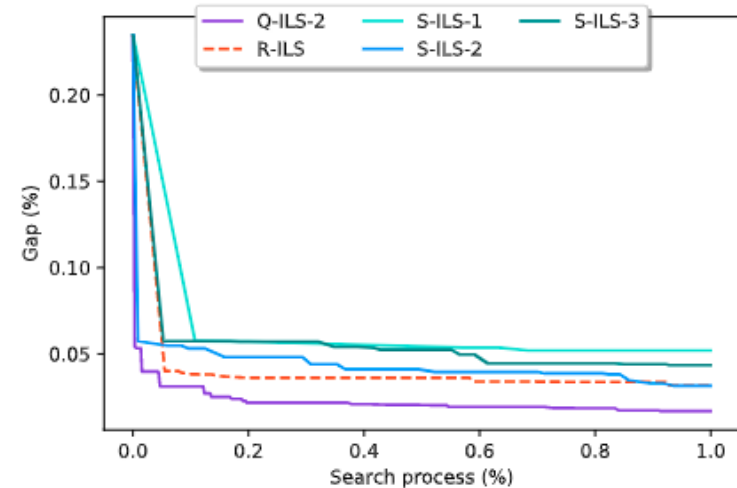
- QILS finds better solutions than RILS & ILS

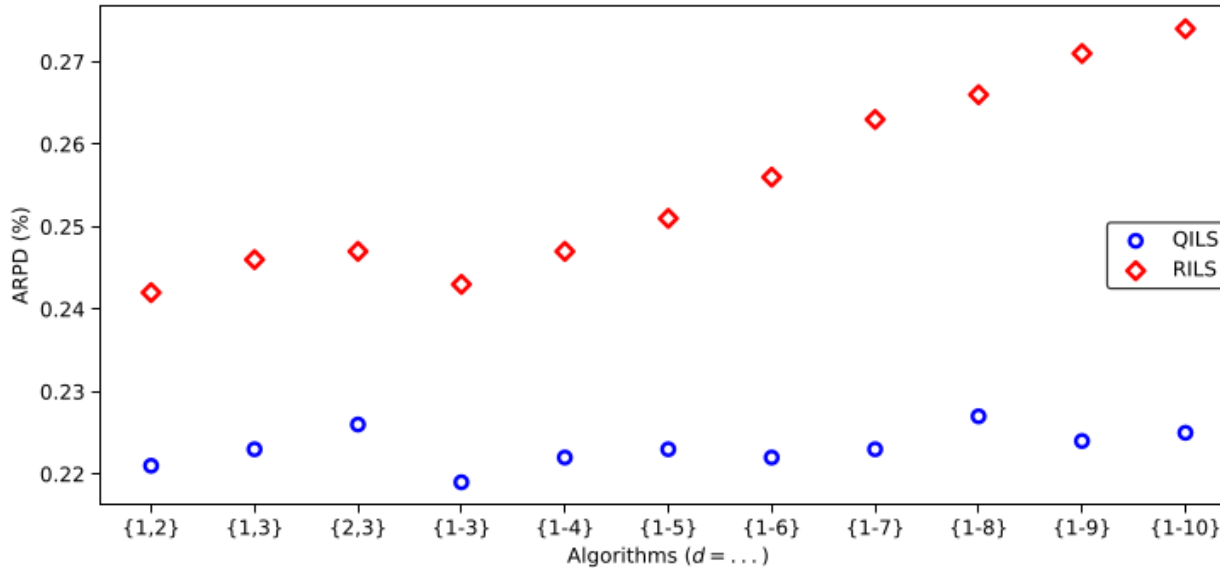
Instance	Algorithms														
	IILS ₁			IILS ₂			IILS ₃			RILS			QILS		
	AvS	BS	T (s)	AvS	BS	T (s)	AvS	BS	T (s)	AvS	BS	T (s)	AvS	BS	T (s)
berlin52	0.204	0	0.610	0.353	0	0.900	0.752	0	0.600	0.164	0	1.100	0	0	0.663
st70	0.993	0	6.700	0.874	0	6.100	1.037	0	7.010	0.800	0	6.400	0.037	0	9.883
kroA100	0.684	0.108	8.200	0.432	0.301	11.500	2.180	0.601	10.400	0.349	0	11.600	0.005	0	10.200
rd100	1.707	0.695	16.200	1.808	0.506	14.600	3.932	3.173	18.900	1.145	0.430	16.800	0.173	0	25.933
lin105	0.869	0	13.300	0.846	0	15.100	2.288	1.015	16.500	0.689	0	13.600	0	0	15.247
pr124	0.491	0.097	24.500	0.281	0	31.800	1.882	0.935	34.500	0.385	0	30.300	0.004	0	34.116
ch130	2.399	1.408	58.300	1.039	0.131	60.500	3.666	3.159	50.700	1.468	0.229	65.600	0.546	0.262	73.380
ch150	1.656	0.490	43.500	1.765	1.103	47.500	4.519	3.676	43.400	1.419	0.322	46.200	0.465	0.077	50.796
u159	1.713	1.440	61.200	1.262	0	66.000	2.548	2.405	70.600	1.702	0	65.100	0.492	0	55.360
d198	0.650	0.234	170.300	0.772	0.190	188.200	2.260	1.141	201.400	0.699	0.057	185.100	0.263	0.165	220.863
kroA200	1.158	0.661	184.400	0.718	0.054	214.100	2.329	1.583	214.100	0.883	0.129	221.900	0.352	0.051	237.510
ts225	0.811	0.187	62.000	0.968	0.696	47.000	2.749	0.187	46.900	0.723	0	46.900	0.013	0	48.030
pr264	1.139	0.092	180.600	1.076	0.263	177.600	5.709	3.446	170.600	1.409	0.794	170.100	0.402	0	207.061
a280	2.443	0.814	314.500	2.869	1.978	347.200	6.126	4.847	333.200	2.559	1.590	362.200	0.587	0	396.827
pr299	2.491	1.836	605.300	2.075	1.828	646.700	5.493	3.779	644.900	2.654	2.139	618.800	0.805	0.151	704.277
lin318	2.049	1.278	591.300	2.153	1.604	605.600	3.048	2.563	632.500	2.144	1.549	604.900	1.559	0.895	688.655
fl417	2.712	1.113	683.700	2.900	1.846	701.900	7.984	5.843	706.300	2.378	1.374	738.900	0.966	0.430	796.136
pr439	4.406	3.553	709.300	4.029	3.622	690.700	6.281	5.441	615.500	3.848	2.174	722.100	2.308	0.755	778.177
pcb442	2.932	2.030	681.600	2.992	1.798	661.300	5.577	4.984	687.500	2.457	1.587	698.000	1.568	1.061	758.390
d493	3.020	1.617	594.900	3.657	2.603	648.400	7.051	5.483	604.200	3.007	2.311	659.500	2.135	1.451	643.362
vm1084	5.266	3.703	2186.200	5.398	4.217	1953.125	12.399	12.112	2006.400	5.082	3.610	2082.000	4.217	3.025	2012.244
d1291	5.177	3.335	1946.600	5.658	3.903	1817.900	10.422	10.090	1911.900	5.498	2.248	1808.100	4.295	3.173	1835.273
u1817	6.278	5.409	596.500	7.812	6.250	667.900	10.801	10.281	771.841	7.153	4.472	765.800	4.858	4.142	664.385
u2152	6.162	4.825	778.300	7.416	6.001	880.000	11.447	10.701	956.400	7.412	4.493	850.500	4.841	3.836	919.772

- QILS finds better solutions than RILS & ILS

Instance	Algorithms														
	IILS ₁			IILS ₂			IILS ₃			RILS			QILS		
	AvS	BS	T (s)	AvS	BS	T (s)	AvS	BS	T (s)	AvS	BS	T (s)	AvS	BS	T (s)
berlin52	0.167	0	0.7	0.239	0	0.6	0.219	0	0.7	0.003	0.000	0.2	0	0	0.2
st70	0.770	0.148	4.2	0.415	0	4.1	0.815	0.148	3.1	0.681	0.148	3.2	0.109	0	4.5
kroA100	0.583	0.000	7.7	0.328	0	8.1	0.438	0	8.7	0.227	0	8.1	0	0	7.3
rd100	0.885	0.695	11.1	0.927	0	9.8	1.088	0	10.2	0.714	0	10.6	0.201	0	14.2
lin105	0.977	0	7.8	0.759	0	9.3	0.462	0.000	8.7	0.174	0	5.6	0	0	4.5
pr124	0.571	0	11.9	0.377	0	10.7	0.469	0.078	12.3	0.221	0	8.9	0	0	8.7
ch130	1.434	0.802	27.4	1.126	0.245	24.4	1.198	0.769	26.6	1.129	0.720	25.2	0.359	0	40.4
ch150	2.106	1.486	23.8	1.385	0.475	24.6	1.544	0.659	27.0	0.928	0.414	19.6	0.374	0	33.2
u159	1.435	0.803	44.2	0.586	0.000	42.8	0.507	0.000	34.1	0.264	0.000	33.1	0.112	0	42.9
d198	1.363	0.976	122.8	0.887	0.253	121.3	0.749	0.418	106.5	0.368	0.228	97.4	0.180	0.057	147.5
kroA200	1.282	0.855	153.4	0.420	0.000	148.3	0.719	0.320	157.2	0.344	0.000	211.6	0.150	0.000	194.8
ts225	1.043	0.641	84.1	0.585	0.285	95.2	0.531	0.364	108.1	0.077	0.000	95.7	0.002	0	108.9
pr264	1.257	0.995	107.7	0.796	0.311	101.4	0.500	0.033	112.9	0.390	0.033	120.8	0.172	0	135.8
a280	4.044	3.024	216.3	2.497	0.310	214.4	2.389	0.698	221.2	2.212	0.388	266.2	0.498	0	276.5
pr299	3.741	2.706	403.1	2.336	0.689	400.7	2.557	1.384	380.8	2.495	1.600	451.2	0.274	0	440.7
lin318	3.755	2.927	407.3	2.626	1.394	425.1	2.527	1.554	403.0	2.225	1.651	492.1	0.795	0.302	479.6
rd400	4.651	3.802	475.7	3.432	2.375	486.0	3.491	1.983	475.0	2.927	0.962	541.5	1.461	0.825	519.4
fl417	4.235	1.425	457.9	2.535	1.239	467.1	2.332	0.995	422.8	1.514	0.590	448.4	0.339	0.211	553.0
pr439	3.157	1.791	602.8	3.215	0.889	501.2	3.287	1.034	499.5	3.057	1.959	640.2	1.392	0.438	528.2
pcb442	3.313	2.578	616.6	3.161	2.407	502.6	3.475	2.724	511.7	2.385	1.804	614.7	1.294	0.640	521.6
d493	4.511	3.437	527.4	3.709	2.437	511.8	3.588	2.774	537.3	2.776	1.703	604.6	1.499	0.923	587.8
vm1084	5.887	4.984	2073.4	5.172	4.011	2039.7	5.675	4.271	1845.6	4.911	3.458	1822.8	3.717	3.061	2608.3
d1291	6.555	4.092	2356.5	5.799	3.870	1605.3	6.129	4.335	1541.2	5.547	2.327	1580.7	3.418	2.281	2060.1
u1817	7.685	6.250	1232.7	7.220	6.087	1323.2	7.354	6.250	1488.8	6.003	3.844	942.1	4.366	3.449	1416.0
u2152	7.008	6.001	1169.3	7.421	6.001	1487.9	7.378	6.001	1306.0	6.357	4.123	1261.3	4.531	3.947	1022.3

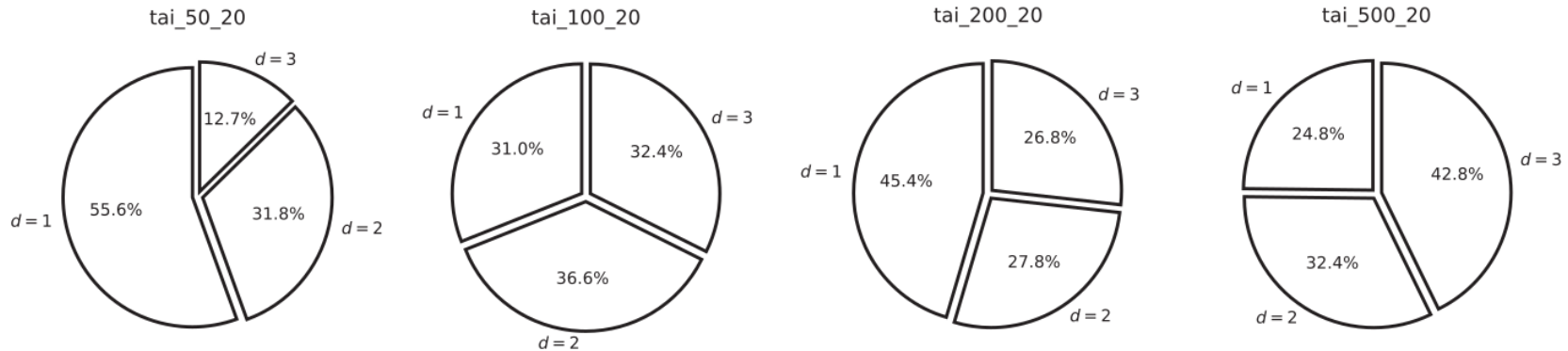
- ▶ QILS finds better solutions than RILS & ILS
- ▶ QILS convergences faster toward better solutions
- ▶ Not really competitive to the state-of-the-art





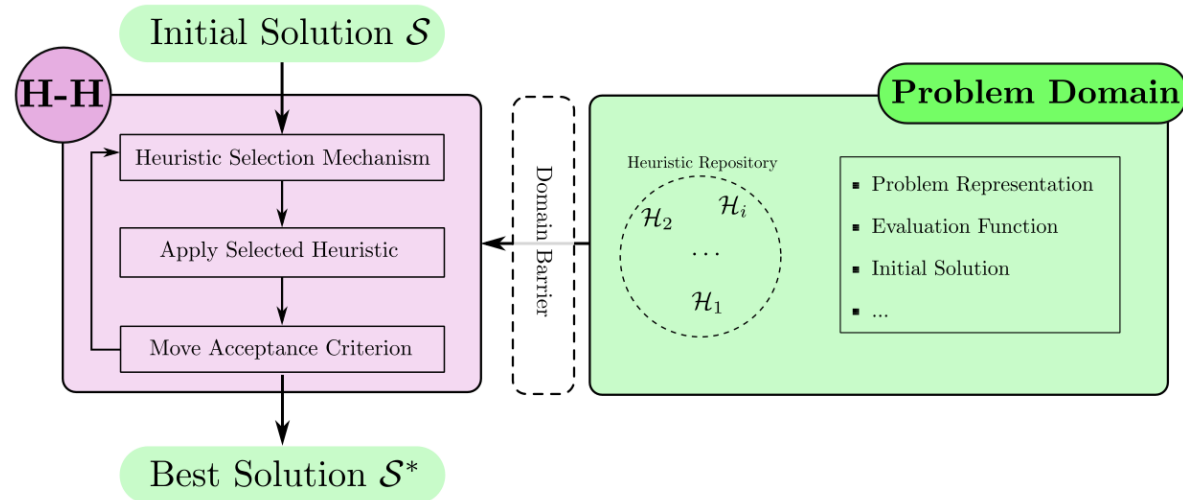
4. Adaptiveness of QILS

- Contribution of operators to the total improvement



Generalizability of the proposed framework

- Hyper-heuristic: a high-level automated search methodology that automates design/selection of heuristics



Generalizability of the proposed framework

- ▶ Comparison to 20 competitors of CheSC 2011 and three state-of-the-art:
 - MSHH: A multi-stage hyper-heuristic proposed by Kheiri & Özcan (2016);
 - QHH: A reinforcement learning based hyper-heuristic proposed by Choong et al. (2018);
 - Modified CF-AM: An improved choice function heuristic proposed by

- ▶ Problem domains:
 - SAT, Bin packing, PFSP, TSP, VRP (5 instances for each)

Generalizability of the proposed framework

Rank	Algorithm	Score for each domain					Total score
		SAT	BP	PFSP	TSP	VRP	
1	SSHH	47.0	38.0	23.5	38.0	12.0	158.5
2	AdapHH	18.7	44.0	27.0	29.5	10.0	129.2
3	QHH	26.5	12.0	28.0	34.0	25.0	125.5
4	QILS	29.5	6.0	30.5	15.0	8.0	89.0
5	ML	3.8	6.0	29.5	5.3	18.0	62.7
6	VNS-TW	19.5	1.0	21.0	6.5	5.0	53.0
7	NAHH	8.0	16.0	16.5	8.0	4.0	52.5
8	PHUNTER	2.8	2.0	2.0	14.5	28.0	49.3
9	EPH	0.0	5.0	9.0	22.5	9.0	45.5
10	KSATS-HH	11.7	6.0	0.0	0.0	17.0	34.7

When numerous operators are available

- ▶ Determine a subset of efficient operators
 - How many & which operators to include in the portfolio?
 - Very time-consuming pre-processing phase to assess operators
 - Ignores the fact "the behavior of operators change depending on the region of search space"